

Kinora Thoracic Spine & Back Clinical AI Orientation (Part 3)

Structured reference for RAG / AI-assisted physiotherapy consultation

Thoracic Spine, Ribs, Breathing, Scapula, TOS, Fascia & Myofascial Chains

Version 1.0 — Kinora Admin Conocimientos Upload

Disclaimer

This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed medical/physiotherapy care. Content reflects established musculoskeletal medicine and sports rehabilitation concepts but must be verified against current peer-reviewed literature, local protocols, and individual patient presentation. Always screen for red flags requiring urgent medical referral (cardiac, pulmonary, cauda equina, progressive neurological deficit, infection, malignancy). Diagnostic sensitivity/specificity values are approximate and vary by study.

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1. Thoracic Spine / Regional Structures

RECORD: Vertebral Level T1

ID: VERT-T-01

Vertebral Level: T1

Region: Upper thoracic (cervicothoracic transition zone)

Bone: Body smaller than T2; prominent transverse processes with costal facets for rib 1; no body demifacets for rib (full facet on T1); vertebral foramen smaller than cervical

Facet Orientation: Superior facets face posterior-superior; inferior facets posterior-inferior; more sagittal than mid-thoracic (~45-55 deg coronal); cervicothoracic transition

Disc Characteristics: T1-T2 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, ligamentum flavum, intertransverse, interspinous; costovertebral/costotransverse caps for rib 1; supraspinous ligament continuous with nuchal ligament superiorly

Muscles: Semispinalis cervicis/thoracis, multifidus, rotatores, longissimus, iliocostalis, levator scapulae (TP), splenius attachments; deep neck extensors at cervicothoracic junction

Innervation: Segmental dorsal rami T1 (major contributor to T1 radiculopathy/brachial plexus overlap); ventral rami T1 form lower trunk/brachial plexus components

Blood Supply: Segmental arteries from supreme intercostal/vertebral tributaries; vertebral venous plexus; first intercostal artery branch patterns

Biomechanics: Minimal independent flexion-extension; coupled with C7-T2; rib 1 ring creates stiff segment; critical for scapular rhythm and TOS proximity

Normal ROM: Flexion 3-5 deg, extension 2-4 deg, lateral flexion 3-5 deg, rotation 2-4 deg segmentally; region moves more as cervicothoracic unit

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 1 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T1 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: T1 radiculopathy, cervicothoracic dysfunction, first rib hypomobility, TOS-related pain referral, Scheuermann apex nearby in adolescents

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T2

ID: VERT-T-02

Vertebral Level: T2

Region: Upper thoracic (cervicothoracic transition zone)

Bone: T2 body with paired superior/inferior costal demifacets for rib 2; pedicles, laminae, spinous process angled inferiorly; transverse costal facet for rib 2 tubercle

Facet Orientation: Zygapophyseal facets oriented ~55-65 deg to coronal plane; T2 superior/inferior facets permit rotation with limited flexion-extension

Disc Characteristics: T2-T3 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, ligamentum flavum, interspinous, supraspinous, costovertebral and costotransverse ligaments for rib 2; radiate ligaments stabilize costovertebral joint

Muscles: Multifidus, rotatores, semispinalis thoracis, longissimus thoracis, iliocostalis, rhomboids (upper), trapezius (middle), serratus posterior superior

Innervation: T2 medial branch dorsal rami (facet/facet capsule); ventral rami T2 intercostal nerve segment

Blood Supply: Segmental artery T2; dorsal branch supplies facet; intercostal artery from aorta segment T2

Biomechanics: Rib 2 ring increases stiffness; upper thoracic contributes to scapular base stability; rotation primary motion with respiration

Normal ROM: Flexion 2-4 deg, extension 2-3 deg, lateral flexion 2-4 deg, rotation 2-4 deg per segment

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 2 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T2 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Upper thoracic hypomobility, costovertebral dysfunction rib 2, postural kyphosis segments, facet syndrome T2-T3

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T3

ID: VERT-T-03

Vertebral Level: T3

Region: Upper thoracic (cervicothoracic transition zone)

Bone: T3 body with paired superior/inferior costal demifacets for rib 3; pedicles, laminae, spinous process angled inferiorly; transverse costal facet for rib 3 tubercle

Facet Orientation: Zygapophyseal facets oriented ~55-65 deg to coronal plane; T3 superior/inferior facets permit rotation with limited flexion-extension

Disc Characteristics: T3-T4 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, ligamentum flavum, interspinous, supraspinous, costovertebral and costotransverse ligaments for rib 3; radiate ligaments stabilize costovertebral joint

Muscles: Multifidus, rotatores, semispinalis thoracis, longissimus thoracis, iliocostalis, rhomboids (upper), trapezius (middle), serratus posterior superior

Innervation: T3 medial branch dorsal rami (facet/facet capsule); ventral rami T3 intercostal nerve segment

Blood Supply: Segmental artery T3; dorsal branch supplies facet; intercostal artery from aorta segment T3

Biomechanics: Rib 3 ring increases stiffness; upper thoracic contributes to scapular base stability; rotation primary motion with respiration

Normal ROM: Flexion 2-4 deg, extension 2-3 deg, lateral flexion 2-4 deg, rotation 2-4 deg per segment

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 3 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T3 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Upper thoracic hypomobility, costovertebral dysfunction rib 3, postural kyphosis segments, facet syndrome T3-T4

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T4

ID: VERT-T-04

Vertebral Level: T4

Region: Upper thoracic (cervicothoracic transition zone)

Bone: T4 body with paired superior/inferior costal demifacets for rib 4; pedicles, laminae, spinous process angled inferiorly; transverse costal facet for rib 4 tubercle

Facet Orientation: Zygapophyseal facets oriented ~55-65 deg to coronal plane; T4 superior/inferior facets permit rotation with limited flexion-extension

Disc Characteristics: T4-T5 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, ligamentum flavum, interspinous, supraspinous, costovertebral and costotransverse ligaments for rib 4; radiate ligaments stabilize costovertebral joint

Muscles: Multifidus, rotatores, semispinalis thoracis, longissimus thoracis, iliocostalis, rhomboids (upper), trapezius (middle), serratus posterior superior

Innervation: T4 medial branch dorsal rami (facet/facet capsule); ventral rami T4 intercostal nerve segment

Blood Supply: Segmental artery T4; dorsal branch supplies facet; intercostal artery from aorta segment T4

Biomechanics: Rib 4 ring increases stiffness; upper thoracic contributes to scapular base stability; rotation primary motion with respiration

Normal ROM: Flexion 2-4 deg, extension 2-3 deg, lateral flexion 2-4 deg, rotation 2-4 deg per segment

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 4 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T4 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Upper thoracic hypomobility, costovertebral dysfunction rib 4, postural kyphosis segments, facet syndrome T4-T5

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T5

ID: VERT-T-05

Vertebral Level: T5

Region: Mid thoracic (typical thoracic cage)

Bone: T5 typical thoracic vertebral body; heart/ mediastinum anterior; spinous process long and overlapping; rib 5 articulations bilateral

Facet Orientation: Classic thoracic facet ~60 deg coronal; T5 resists flexion-extension, favors rotation and lateral flexion within rib cage constraints

Disc Characteristics: T5-T6 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum, interspinous, supraspinous; costovertebral joint capsule; costotransverse; superior/inferior costotransverse ligaments

Muscles: Multifidus, rotatores (long segment), semispinalis, erector spinae, intercostals, serratus posterior inferior (lower mid), latissimus origin broad

Innervation: T5 dorsal ramus medial branch; intercostal nerve T5 (ventral ramus); lateral cutaneous branch at mid-axillary line

Blood Supply: Intercostal artery and vein pair T5; segmental spinal branches to cord and meninges

Biomechanics: Mid-thoracic cage is most rigid; torsion transmitted during gait; scoliosis apex often T8-T9 region in adolescents

Normal ROM: Flexion 2-3 deg, extension 2 deg, lateral flexion 2-3 deg, rotation 2-3 deg segmentally

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 5 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T5 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Scheuermann kyphosis apex, idiopathic scoliosis, compression fracture (osteoporosis), intercostal neuralgia, postural pain

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T6

ID: VERT-T-06

Vertebral Level: T6

Region: Mid thoracic (typical thoracic cage)

Bone: T6 typical thoracic vertebral body; heart/ mediastinum anterior; spinous process long and overlapping; rib 6 articulations bilateral

Facet Orientation: Classic thoracic facet ~60 deg coronal; T6 resists flexion-extension, favors rotation and lateral flexion within rib cage constraints

Disc Characteristics: T6-T7 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum, interspinous, supraspinous; costovertebral joint capsule; costotransverse; superior/inferior costotransverse ligaments

Muscles: Multifidus, rotatores (long segment), semispinalis, erector spinae, intercostals, serratus posterior inferior (lower mid), latissimus origin broad

Innervation: T6 dorsal ramus medial branch; intercostal nerve T6 (ventral ramus); lateral cutaneous branch at mid-axillary line

Blood Supply: Intercostal artery and vein pair T6; segmental spinal branches to cord and meninges

Biomechanics: Mid-thoracic cage is most rigid; torsion transmitted during gait; scoliosis apex often T8-T9 region in adolescents

Normal ROM: Flexion 2-3 deg, extension 2 deg, lateral flexion 2-3 deg, rotation 2-3 deg segmentally

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 6 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T6 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Scheuermann kyphosis apex, idiopathic scoliosis, compression fracture (osteoporosis), intercostal neuralgia, postural pain

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T7

ID: VERT-T-07

Vertebral Level: T7

Region: Mid thoracic (typical thoracic cage)

Bone: T7 typical thoracic vertebral body; heart/ mediastinum anterior; spinous process long and overlapping; rib 7 articulations bilateral

Facet Orientation: Classic thoracic facet ~60 deg coronal; T7 resists flexion-extension, favors rotation and lateral flexion within rib cage constraints

Disc Characteristics: T7-T8 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum, interspinous, supraspinous; costovertebral joint capsule; costotransverse; superior/inferior costotransverse ligaments

Muscles: Multifidus, rotatores (long segment), semispinalis, erector spinae, intercostals, serratus posterior inferior (lower mid), latissimus origin broad

Innervation: T7 dorsal ramus medial branch; intercostal nerve T7 (ventral ramus); lateral cutaneous branch at mid-axillary line

Blood Supply: Intercostal artery and vein pair T7; segmental spinal branches to cord and meninges

Biomechanics: Mid-thoracic cage is most rigid; torsion transmitted during gait; scoliosis apex often T8-T9 region in adolescents

Normal ROM: Flexion 2-3 deg, extension 2 deg, lateral flexion 2-3 deg, rotation 2-3 deg segmentally

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 7 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T7 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Scheuermann kyphosis apex, idiopathic scoliosis, compression fracture (osteoporosis), intercostal neuralgia, postural pain

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T8

ID: VERT-T-08

Vertebral Level: T8

Region: Mid thoracic (typical thoracic cage)

Bone: T8 typical thoracic vertebral body; heart/ mediastinum anterior; spinous process long and overlapping; rib 8 articulations bilateral

Facet Orientation: Classic thoracic facet ~60 deg coronal; T8 resists flexion-extension, favors rotation and lateral flexion within rib cage constraints

Disc Characteristics: T8-T9 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum, interspinous, supraspinous; costovertebral joint capsule; costotransverse; superior/inferior costotransverse ligaments

Muscles: Multifidus, rotatores (long segment), semispinalis, erector spinae, intercostals, serratus posterior inferior (lower mid), latissimus origin broad

Innervation: T8 dorsal ramus medial branch; intercostal nerve T8 (ventral ramus); lateral cutaneous branch at mid-axillary line

Blood Supply: Intercostal artery and vein pair T8; segmental spinal branches to cord and meninges

Biomechanics: Mid-thoracic cage is most rigid; torsion transmitted during gait; scoliosis apex often T8-T9 region in adolescents

Normal ROM: Flexion 2-3 deg, extension 2 deg, lateral flexion 2-3 deg, rotation 2-3 deg segmentally

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 8 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T8 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Scheuermann kyphosis apex, idiopathic scoliosis, compression fracture (osteoporosis), intercostal neuralgia, postural pain

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T9

ID: VERT-T-09

Vertebral Level: T9

Region: Lower thoracic (thoracolumbar transition zone)

Bone: T9 body larger; rib 9 false rib (costal cartilage joins cartilage above); reduced bony rib ring constraint vs true ribs

Facet Orientation: Lower thoracic facets progressively more sagittal; T9 permits increased flexion-extension vs mid-thoracic

Disc Characteristics: T9-T10 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum; costovertebral for rib 9; thoracolumbar fascia attachments increase inferiorly

Muscles: Multifidus, erector spinae (iliocostalis lumborum transition), QL proximity T12, latissimus, internal/external oblique attachments via fascia

Innervation: T9 segmental nerve; subcostal overlap at T12 region when applicable

Blood Supply: Segmental T9 intercostal/lumbar anastomotic branches

Biomechanics: Reduced rib cage constraint allows more segmental flexion; important for lifting mechanics and thoracolumbar load transfer

Normal ROM: Flexion 4-6 deg, extension 3-5 deg, lateral flexion 3-4 deg, rotation 2-4 deg

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 9 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T9 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Thoracolumbar strain, facet irritation, disc degeneration (uncommon), rib cartilage pain referral

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation palflo press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T10

ID: VERT-T-10

Vertebral Level: T10

Region: Lower thoracic (thoracolumbar transition zone)

Bone: T10 body larger; rib 10 false rib (costal cartilage joins cartilage above); reduced bony rib ring constraint vs true ribs

Facet Orientation: Lower thoracic facets progressively more sagittal; T10 permits increased flexion-extension vs mid-thoracic

Disc Characteristics: T10-T11 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum; costovertebral for rib 10; thoracolumbar fascia attachments increase inferiorly

Muscles: Multifidus, erector spinae (iliocostalis lumborum transition), QL proximity T12, latissimus, internal/external oblique attachments via fascia

Innervation: T10 segmental nerve; subcostal overlap at T12 region when applicable

Blood Supply: Segmental T10 intercostal/lumbar anastomotic branches

Biomechanics: Reduced rib cage constraint allows more segmental flexion; important for lifting mechanics and thoracolumbar load transfer

Normal ROM: Flexion 4-6 deg, extension 3-5 deg, lateral flexion 3-4 deg, rotation 2-4 deg

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 10 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T10 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Thoracolumbar strain, facet irritation, disc degeneration (uncommon), rib cartilage pain referral

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T11

ID: VERT-T-11

Vertebral Level: T11

Region: Lower thoracic (thoracolumbar transition zone)

Bone: T11 body larger; rib 11 false rib (costal cartilage joins cartilage above); reduced bony rib ring constraint vs true ribs

Facet Orientation: Lower thoracic facets progressively more sagittal; T11 permits increased flexion-extension vs mid-thoracic

Disc Characteristics: T11-T12 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, flavum; costovertebral for rib 11; thoracolumbar fascia attachments increase inferiorly

Muscles: Multifidus, erector spinae (iliocostalis lumborum transition), QL proximity T12, latissimus, internal/external oblique attachments via fascia

Innervation: T11 segmental nerve; subcostal overlap at T12 region when applicable

Blood Supply: Segmental T11 intercostal/lumbar anastomotic branches

Biomechanics: Reduced rib cage constraint allows more segmental flexion; important for lifting mechanics and thoracolumbar load transfer

Normal ROM: Flexion 4-6 deg, extension 3-5 deg, lateral flexion 3-4 deg, rotation 2-4 deg

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 11 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T11 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Thoracolumbar strain, facet irritation, disc degeneration (uncommon), rib cartilage pain referral

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Vertebral Level T12

ID: VERT-T-12

Vertebral Level: T12

Region: Lower thoracic (thoracolumbar transition zone)

Bone: Body resembles lumbar vertebra; articular processes large; often no costal facets on body (floating rib 12); mamillary processes may appear

Facet Orientation: Facets transition toward lumbar orientation (~90 deg sagittal); inferior facets face lateral; superior facets medial; increased sagittal plane freedom

Disc Characteristics: T12-TL1 disc: nucleus pulposus constrained by rib cage; annulus fibrosus thinner than lumbar; limited disc height change with loading; posterior annulus reinforced; lateral bulge less common than lumbar due to rigidity; Schmorl nodes possible in adolescents at endplates

Ligaments: ALL, PLL, ligamentum flavum; thoracolumbar fascia attachment; iliolumbar ligament analogues at transition; strong interspinous region

Muscles: Multifidus, erector spinae, quadratus lumborum proximity, latissimus dorsi via TL fascia, psoas minor attachment region

Innervation: T12 dorsal rami; ventral rami contribute to subcostal nerve; overlap with L1 innervation patterns at transition

Blood Supply: Segmental T12 arteries from lumbar/intercostal anastomoses; Batson plexus drainage

Biomechanics: Transition zone: greatest thoracolumbar shear potential; facet orientation permits more sagittal motion than mid-thoracic

Normal ROM: Flexion 6-8 deg, extension 5-7 deg, lateral flexion 4-6 deg, rotation 3-5 deg; highest thoracic segmental ROM

Arthrokinematics: Facet glides: extension — inferior facet slides inferior-posterior on superior facet of segment below; flexion reverses; rotation couples with contralateral facet opening; costovertebral joint rolls/slides with respiration and trunk rotation

Osteokinematics: Regional flexion-extension minimal mid-thoracic; rotation and lateral flexion coupled; whole thoracic kyphosis 20-40 deg; upper and lower transitions contribute disproportionately to regional motion

Force Transmission: Axial compression through vertebral bodies and discs; lateral rib cage dissipates forces; facet joints share shear in rotation; costovertebral joints transfer bucket/pump handle forces during breathing

Shock Absorption: Disc viscoelasticity limited vs lumbar; rib cage hoop tension; paraspinal muscle eccentric control; kyphotic curve distributes axial load anteriorly through vertebral bodies

Respiratory Function: Rib 12 attachment contributes to thoracic volume change; segmental rotation with inspiration; posterior intercostals and paraspinals stabilize during forced breathing

Clinical Examination: Observation: kyphosis, scoliosis, winging; active ROM thoracic rotation/lateral flexion in sitting; passive accessory glide PA/central; neuro screen dermatomes T1-T12 (T4 nipple, T10 umbilicus landmarks)

Palpation: Spinous process T12 midline; transverse processes ~3 cm lateral; paraspinal soft tissue; costotransverse angle palpation for rib hypomobility; facet tenderness 1-2 cm lateral to SP

Segmental Mobility: PA central/over-unilateral glides; spring test for stiffness; compare segment above/below; rib spring at costotransverse angle; note pain reproduction vs stiffness only

Special Tests: Slump/neural tension if radicular; Adam's forward bend for scoliosis; thoracic extension rotation test; first rib mobility test if T1; instability tests generally not applicable except thoracolumbar transition

Common Dysfunctions: Segmental hypomobility, hypermobility at transition zones, postural extension loss, facet irritation, costovertebral restriction, muscle guarding pattern

Common Pathologies: Thoracolumbar junction syndrome, pars stress at transition, disc herniation (rare but possible), facet arthropathy, QL referral overlap

Imaging: X-ray AP/lateral for alignment, fractures, scoliosis; CT for complex fractures; MRI for disc, cord, tumor, infection; bone scan for metastases in oncology screen

Manual Therapy: Grade I-II mobilization for pain; grade III-IV PA glides for hypomobility; costovertebral mobilization; muscle energy for rotation; HVLA only with clear diagnosis and consent

Rehabilitation: Phase 1: pain control, breathing, postural awareness; Phase 2: mobility thoracic rotation/extension; Phase 3: scapular-thoracic integration; Phase 4: sport-specific rotation/load

Mobility Exercises: Open book, thread the needle, foam roller extension (mid-thoracic), cat-camel segmental, seated rotation with block, rib expansion breathing in side-lying

Strengthening: Prone extension holds, row variations, face pulls, dead bug with rotation control, anti-rotation pallof press, bird dog, lower trap activation

Return to Sport: Full pain-free ROM; symmetric rotation >45 deg each side; no neural symptoms; progressive loaded rotation and extension; contact sports after fracture healing 8-12 weeks if applicable

Clinical Guidelines: NICE low back/thoracic pain: reassurance, activity modification, manual therapy optional; red flags: cauda equina, progressive neuro deficit, fever, trauma, cancer history

Evidence: Thoracic manipulation may reduce neck pain (indirect evidence); motor control training for scapular dyskinesis Level B; see Cochrane reviews spinal manipulative therapy for mixed chronic pain

References: Neumann Kinesiology of the Musculoskeletal System; Bogduk clinical anatomy; ISIS thoracic spine guidelines; PubMed: thoracic spine mobilization systematic reviews

RECORD: Thoracic Facet Joints (Zygapophyseal)

ID: JOINT-T-FACET

Vertebral Level: T1-T12 bilateral pairs

Region: Posterior thoracic column

Bone: Inferior articular facet of cranial vertebra with superior facet of caudal vertebra

Facet Orientation: ~60 degrees to coronal plane (upper slightly more sagittal, lower trending sagittal at T11-T12)

Disc Characteristics: N/A — facet joint synovial; interposed meniscoids/fat pads

Ligaments: Joint capsule, flavum contribution posteriorly, medial branch nerve innervates capsule

Muscles: Multifidus, rotatores, semispinalis — dynamic stabilization; erector spinae global

Innervation: Medial branch dorsal ramus each level; facet pain maps locally and referred paraspinal

Blood Supply: Segmental arterial anastomoses from dorsal branches

Biomechanics: Guide rotation; resist anterior shear in extension; load sharing in axial compression ~15-20% in lumbar analog

Normal ROM: Primary plane rotation; flexion-extension limited 2-5 deg per segment

Arthrokinematics: Convex-on-concave glides with coupled motion; capsular slack in flexion, taut in extension

Osteokinematics: Thoracic rotation 35-50 deg total; lateral flexion 25-35 deg total

Force Transmission: Posterior column shear and torsion; facet tropism rare in thoracic vs lumbar

Shock Absorption: Minimal; facet loading increases in extension and rotation end-range

Respiratory Function: Indirect — rib cage motion alters facet loading with breathing

Clinical Examination: Extension pain suggests facet loading; rotation restriction; palpation 1-2 cm lateral to SP

Palpation: Facet line parallel to SP; tenderness grading; compare bilateral

Segmental Mobility: PA glide, gapping, rotation mobilization

Special Tests: No single gold standard; extension-rotation provocative; medial branch block diagnostic for chronic facet pain

Common Dysfunctions: Facet arthropathy, synovial impingement, meniscoid entrapment, post-whiplash upper thoracic

Common Pathologies: Facet joint syndrome, osteoarthritis, inflammatory spondyloarthropathy

Imaging: X-ray limited; CT/MRI for effusion, synovitis, fracture; bone scan for active facet arthropathy

Manual Therapy: Mobilization grades III-IV, manipulation when appropriate, flexion distraction rare in thoracic

Rehabilitation: Extension control exercises, rotation mobility, avoid repeated end-range extension loading early

Mobility Exercises: Open book, seated rotation, extension in neutral (not hyperextension)

Strengthening: Anti-extension core, multifidus activation, rows with scapular retraction

Return to Sport: Pain-free rotation and controlled extension under load

Clinical Guidelines: Diagnostic medial branch blocks for chronic facet-mediated pain prior to radiofrequency

Evidence: Thoracic facet pain less studied than lumbar; medial branch RF evidence extrapolated from lumbar literature

References: Bogduk facet joint innervation; Cohen SP facet interventions systematic reviews

RECORD: Thoracic Intervertebral Discs

ID: JOINT-T-DISC

Vertebral Level: T1-T12 intervertebral

Region: Anterior thoracic column between vertebral bodies

Bone: Endplates of adjacent vertebral bodies

Facet Orientation: N/A

Disc Characteristics: Thin relative to lumbar; nucleus constrained; annulus lamellae; limited hydration change; Schmorl nodes in adolescents; herniation uncommon (~0.25-4% of disc herniations)

Ligaments: ALL anterior, PLL posterior, endplate cartilage

Muscles: Indirect — erector spinae tension affects loading

Innervation: Sinuvertebral nerve (recurrent meningeal); outer annulus only; referred pain to chest wall possible

Blood Supply: Endplate diffusion nutrition; segmental vessels peripherally

Biomechanics: Resists compression; minimal flexion-extension contribution due to rib cage

Normal ROM: Disc contributes to micro-motion; macro ROM limited by ribs

Arthrokinematics: Nucleus migrates slightly with flexion-extension in free segments

Osteokinematics: Disc height loss with aging/degeneration less clinically prominent than lumbar

Force Transmission: Primary axial load bearer anterior column

Shock Absorption: Viscoelastic nucleus and annulus; rib cage shares load

Respiratory Function: Minimal direct; indirect via thoracic cage expansion

Clinical Examination: Central PA pain; flexion may provoke; dermatomal symptoms if herniation (rare)

Palpation: Not directly palpable; indirect via segmental tenderness

Segmental Mobility: Central PA spring; flexion opening

Special Tests: Slump test if radicular; Valsalva if discogenic radicular suspected

Common Dysfunctions: Internal disc disruption (rare thoracic), endplate edema Modic changes

Common Pathologies: Disc herniation, Scheuermann apophyseal ring, infection (discitis), tumor

Imaging: MRI gold standard; T2 signal loss degeneration; contrast for infection/tumor

Manual Therapy: Gentle mobilization; avoid aggressive flexion in acute radicular presentation
Rehabilitation: McKenzie extension if centralization (case dependent); core stability; gradual loading
Mobility Exercises: Cat-camel gentle range, rotation within tolerance
Strengthening: Progressive extension endurance, row patterns
Return to Sport: No radicular symptoms; MRI clearance if herniation; gradual axial loading
Clinical Guidelines: Thoracic disc herniation: conservative trial unless myelopathy; surgery for cord compression
Evidence: Surgical outcomes for thoracic disc herniation variable; conservative evidence limited case series
References: Wood KB thoracic disc herniation review; PubMed thoracic disc management

RECORD: Costovertebral Joints

ID: JOINT-T-CV

Vertebral Level: T1-T10 typically (ribs 1-10 heads)

Region: Lateral vertebral body — rib head articulation

Bone: Rib head two facets (demifacets on two vertebrae except rib 1, 11, 12 variants)

Facet Orientation: Synovial planar/biplanar joint

Disc Characteristics: N/A

Ligaments: Radiate ligament (stout), intra-articular ligament rib 2-9 often, joint capsule

Muscles: Intercostals, levator costae, serratus posterior

Innervation: Intercostal nerve; joint referral to anterior chest wall

Blood Supply: Intercostal artery branches

Biomechanics: Rib elevation/depression; rotation of rib during respiration; stabilizes rib cage ring

Normal ROM: Small amplitude glides with breathing; larger with forced respiration and trunk motion

Arthrokinematics: Roll-glide with pump/bucket handle rib motion

Osteokinematics: Rib rises with inspiration, descends expiration

Force Transmission: Transfers lateral thoracic loads; coughing/Valsalva increases compressive load

Shock Absorption: Cartilaginous interface; ligamentous give

Respiratory Function: Primary — each costovertebral joint contributes to thoracic volume change

Clinical Examination: Pain with deep breath, cough, rotation; palpation costotransverse angle

Palpation: Posterior angle tenderness; spring test hypomobility

Segmental Mobility: Posteroanterior rib mobilization, muscle energy inhalation/exhalation

Special Tests: Hooking maneuver, rib spring test, reproducing with cough

Common Dysfunctions: Rib fixation, post-traumatic hypomobility, post-surgical pain

Common Pathologies: Costovertebral sprain, inflammatory arthritis, Tietze adjacent referral

Imaging: X-ray rarely shows; CT for fracture/dislocation; MRI for inflammation

Manual Therapy: Mulligan/snag rib elevation, MET, HVLA rib adjustment trained clinicians

Rehabilitation: Breathing retraining, thoracic rotation, intercostal stretching

Mobility Exercises: Side-lying rib expansion, seated rotation with breath

Strengthening: Serratus, intercostal endurance via resisted breathing devices (moderate evidence)

Return to Sport: Pain-free deep breathing and contact without rib pain

Clinical Guidelines: Differentiate cardiac vs musculoskeletal chest pain; red flags cardiac/pulmonary

Evidence: Manual therapy for rib dysfunction mostly expert opinion and case series

References: Greenman principles; manual medicine rib dysfunction texts

RECORD: Costotransverse Joints

ID: JOINT-T-CT

Vertebral Level: T1-T10 transverse process costal facets

Region: Posterolateral — rib tubercle on transverse process

Bone: Rib tubercle articulates with transverse process costal facet

Facet Orientation: Synovial; planar

Disc Characteristics: N/A

Ligaments: Superior/inferior costotransverse ligaments, lateral costotransverse ligament (rib 1)

Muscles: Levator costae brevis/longus, intercostals, rotatores

Innervation: Intercostal nerve dorsal branch region

Blood Supply: Dorsal intercostal branches

Biomechanics: Pivot for rib rotation in transverse plane; stabilizes rib during bucket handle

Normal ROM: Coupled with costovertebral and sternocostal motion

Arthrokinematics: Posterior glide with inspiration at some ribs

Osteokinematics: Rib external rotation during inspiration

Force Transmission: Rotational torque during upper limb loading via serratus/pec minor chain

Shock Absorption: Ligamentous restraint primary

Respiratory Function: Essential for rib cage kinematics

Clinical Examination: TTP at costotransverse angle; pain lifting arm if serratus/rib coupling involved

Palpation: Angle of rib posterior to axilla line; compare left/right spring

Segmental Mobility: Direct rib angle mobilization in side-lying/prone

Special Tests: Rib spring test; seated arm elevation reproducing pain

Common Dysfunctions: Hypomobile rib, slipping rib adjacent structures

Common Pathologies: Costotransverse sprain, stress from coughing, scoliosis asymmetry

Imaging: Usually normal X-ray; ultrasound for dynamic slipping rib

Manual Therapy: Rib raising techniques, MET, soft tissue intercostals

Rehabilitation: Scapular stability, breathing mechanics, gradual loading

Mobility Exercises: Quadruped serratus push, wall slides with expansion

Strengthening: Serratus punch, band pull-apart

Return to Sport: Asymmetric rib pain resolved; full push/pull

Clinical Guidelines: Rule out visceral pathology for unilateral chest wall pain

Evidence: Limited RCTs; clinical tradition strong in manual physiotherapy

References: Manual therapy rib cage literature; Heiderscheit TOS/rib interaction

2. Ribs

RECORD: Rib 1

ID: RIB-01

Rib Number: 1

Type: True rib (vertebrosternal)

Articulations: Single costovertebral facet on T1; costotransverse T1; costochondral junction; no sternocostal demifacet sharing

Muscle Attachments: Scalenus anterior/medius (TOS), subclavius groove, serratus anterior, intercostals, serratus posterior superior

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 1; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T1 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: Shortest broadest rib; protects subclavian vessels/brachial plexus; hypomobility linked to TOS

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Minimal elevation — more caliper than pump due to obliquity

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 1, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Uncommon at rib 1; more TOS/scalene overload

Rib Fracture: Fracture rib 1: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: First rib elevated (cervical rib variant) mimics TOS; not classic slipping rib

Intercostal Strain: Intercostal muscle strain rib 1 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 2

ID: RIB-02

Rib Number: 2

Type: True rib (vertebrosternal)

Articulations: Two demifacets T1-T2; costotransverse T2; costochondral; sternocostal angle prominent

Muscle Attachments: Serratus posterior superior, intercostals, levator scapulae adjacent fascia, rhomboid overlay

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 2; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T2 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: Longest rib; highest bucket handle arc; common post-MVA seatbelt fracture site

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Moderate pump handle at sternocostal junction

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 2, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Sternocostal angle T2 region costochondritis reported

Rib Fracture: Fracture rib 2: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Slipping rib syndrome more common lower ribs but possible 8-10

Intercostal Strain: Intercostal muscle strain rib 2 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 3

ID: RIB-03

Rib Number: 3

Type: True rib (vertebrosternal)

Articulations: Demifacets T2-T3; costotransverse T3; costochondral cartilage; direct sternocostal articulation ribs 1-7

Muscle Attachments: External/internal/innermost intercostals, serratus anterior (ribs 1-8), transversus thoracis inner surface, levator costae

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 3; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T3 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: True rib 3; forms rigid thoracic ring; rotation and elevation in respiration; protects heart/lungs

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Pump handle increases ribs 2-5; rib 3 contributes proportionally

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 3, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondritis at rib 3 costochondral or sternocostal junction; reproducible palpation

Rib Fracture: Fracture rib 3: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Less common true ribs; rule out intercostal neuralgia

Intercostal Strain: Intercostal muscle strain rib 3 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 4

ID: RIB-04

Rib Number: 4

Type: True rib (vertebrosternal)

Articulations: Demifacets T3-T4; costotransverse T4; costochondral cartilage; direct sternocostal articulation ribs 1-7

Muscle Attachments: External/internal/innermost intercostals, serratus anterior (ribs 1-8), transversus thoracis inner surface, levator costae

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 4; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T4 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: True rib 4; forms rigid thoracic ring; rotation and elevation in respiration; protects heart/lungs

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Pump handle increases ribs 2-5; rib 4 contributes proportionally

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 4, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondritis at rib 4 costochondral or sternocostal junction; reproducible palpation

Rib Fracture: Fracture rib 4: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Less common true ribs; rule out intercostal neuralgia

Intercostal Strain: Intercostal muscle strain rib 4 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support
Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early
Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying
Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality
References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 5

ID: RIB-05

Rib Number: 5

Type: True rib (vertebrosternal)

Articulations: Demifacets T4-T5; costotransverse T5; costochondral cartilage; direct sternocostal articulation ribs 1-7

Muscle Attachments: External/internal/innermost intercostals, serratus anterior (ribs 1-8), transversus thoracis inner surface, levator costae

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 5; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T5 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: True rib 5; forms rigid thoracic ring; rotation and elevation in respiration; protects heart/lungs

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Pump handle increases ribs 2-5; rib 5 contributes proportionally

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 5, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondritis at rib 5 costochondral or sternocostal junction; reproducible palpation

Rib Fracture: Fracture rib 5: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Less common true ribs; rule out intercostal neuralgia

Intercostal Strain: Intercostal muscle strain rib 5 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 6

ID: RIB-06

Rib Number: 6

Type: True rib (vertebrosternal)

Articulations: Demifacets T5-T6; costotransverse T6; costochondral cartilage; direct sternocostal articulation ribs 1-7

Muscle Attachments: External/internal/innermost intercostals, serratus anterior (ribs 1-8), transversus thoracis inner surface, levator costae

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 6; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T6 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: True rib 6; forms rigid thoracic ring; rotation and elevation in respiration; protects heart/lungs

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Pump handle increases ribs 2-5; rib 6 contributes proportionally

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 6, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondritis at rib 6 costochondral or sternocostal junction; reproducible palpation

Rib Fracture: Fracture rib 6: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Less common true ribs; rule out intercostal neuralgia

Intercostal Strain: Intercostal muscle strain rib 6 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 7

ID: RIB-07

Rib Number: 7

Type: True rib (vertebrosternal)

Articulations: Demifacets T6-T7; costotransverse T7; costochondral cartilage; direct sternocostal articulation ribs 1-7

Muscle Attachments: External/internal/innermost intercostals, serratus anterior (ribs 1-8), transversus thoracis inner surface, levator costae

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 7; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T7 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: True rib 7; forms rigid thoracic ring; rotation and elevation in respiration; protects heart/lungs

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Pump handle increases ribs 2-5; rib 7 contributes proportionally

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 7, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondritis at rib 7 costochondral or sternocostal junction; reproducible palpation

Rib Fracture: Fracture rib 7: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Less common true ribs; rule out intercostal neuralgia

Intercostal Strain: Intercostal muscle strain rib 7 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 8

ID: RIB-08

Rib Number: 8

Type: False rib (vertebrochondral)

Articulations: Costovertebral T7-T8; costotransverse; costal cartilage joins rib 9 cartilage (shared), not sternum directly

Muscle Attachments: Intercostals, diaphragm attachment lower ribs, transversus abdominis via costal margin, obliques

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 8; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T8 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: False rib 8; greater caliper motion; floating anterior attachment via cartilage cascade

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Reduced direct sternal pump; indirect via cartilage unit ribs 8-10

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 8, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondral junction pain common; slipping rib syndrome ribs 8-10 classic

Rib Fracture: Fracture rib 8: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Rib 8 hypermobile anterior cartilage — Hooking test positive; ultrasound diagnostic

Intercostal Strain: Intercostal muscle strain rib 8 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 9

ID: RIB-09

Rib Number: 9

Type: False rib (vertebrochondral)

Articulations: Costovertebral T8-T9; costotransverse; costal cartilage joins rib 10 cartilage (shared), not sternum directly

Muscle Attachments: Intercostals, diaphragm attachment lower ribs, transversus abdominis via costal margin, obliques

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 9; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T9 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: False rib 9; greater caliper motion; floating anterior attachment via cartilage cascade

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Reduced direct sternal pump; indirect via cartilage unit ribs 8-10

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 9, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondral junction pain common; slipping rib syndrome ribs 8-10 classic

Rib Fracture: Fracture rib 9: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Rib 9 hypermobile anterior cartilage — Hooking test positive; ultrasound diagnostic

Intercostal Strain: Intercostal muscle strain rib 9 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 10

ID: RIB-10

Rib Number: 10

Type: False rib (vertebrochondral)

Articulations: Costovertebral T9-T10; costotransverse; costal cartilage joins rib 11 cartilage (shared), not sternum directly

Muscle Attachments: Intercostals, diaphragm attachment lower ribs, transversus abdominis via costal margin, obliques

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 10; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T10 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: False rib 10; greater caliper motion; floating anterior attachment via cartilage cascade

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Reduced direct sternal pump; indirect via cartilage unit ribs 8-10

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 10, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Costochondral junction pain common; slipping rib syndrome ribs 8-10 classic

Rib Fracture: Fracture rib 10: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Rib 10 hypermobile anterior cartilage — Hooking test positive; ultrasound diagnostic

Intercostal Strain: Intercostal muscle strain rib 10 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 11

ID: RIB-11

Rib Number: 11

Type: Floating rib (vertebral only, no anterior sternal/costal cartilage attachment to sternum)

Articulations: Costovertebral T10-T11; often no costotransverse facet T11-T12 variants; no anterior sternal connection

Muscle Attachments: Quadratus lumborum (rib 12), serratus posterior inferior, iliocostalis lumborum, internal oblique

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 11; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T11 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: Floating rib 11; vulnerable tip trauma; QL attachment rib 12 clinically relevant

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Negligible pump handle

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 11, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Tip tenderness vs renal referral differential lower ribs

Rib Fracture: Fracture rib 11: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Rib 12 fracture more common than slipping; palpate free end

Intercostal Strain: Intercostal muscle strain rib 11 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

RECORD: Rib 12

ID: RIB-12

Rib Number: 12

Type: Floating rib (vertebral only, no anterior sternal/costal cartilage attachment to sternum)

Articulations: Costovertebral T11-T12; often no costotransverse facet T11-T12 variants; no anterior sternal connection

Muscle Attachments: Quadratus lumborum (rib 12), serratus posterior inferior, iliocostalis lumborum, internal oblique

Ligament Attachments: Radiate ligament head; costotransverse superior/inferior/lateral; costochondral periosteum continuity

Blood Supply: Intercostal artery branch 12; anterior intercostal from internal thoracic (upper), musculophrenic (lower)

Innervation: Intercostal nerve T12 ventral ramus; lateral cutaneous branch; anterior cutaneous branch

Biomechanics: Floating rib 12; vulnerable tip trauma; QL attachment rib 12 clinically relevant

Movement During Inspiration: Elevation, external rotation, lateral translation (bucket/caliper/pump components by level)

Movement During Expiration: Depression, internal rotation, passive recoil of cartilage and lung elastic recoil

Pump Handle Motion: Negligible pump handle

Bucket Handle Motion: Transverse plane lateral elevation ribs 6-10 primary; increases AP diameter

Caliper Motion: Lower ribs 8-12 increase transverse thoracic diameter; minimal sternal rise

Clinical Examination: Observe breathing pattern; palpate length of rib; spring test; cough/sneeze repro; neuro screen intercostal

Palpation: Posterior angle rib 12, costochondral junction anterior, compare symmetry

Spring Test: Posterior-anterior pressure on rib angle; assess hypomobility vs hypermobility/pain

Common Injuries: Contusion, stress fracture (rowers, golfers), avulsion in contact sports

Costochondritis: Tip tenderness vs renal referral differential lower ribs

Rib Fracture: Fracture rib 12: point tenderness, crepi, pain inspiration; X-ray often needed; protect breathing; avoid flail if multiple

Slipping Rib Syndrome: Rib 12 fracture more common than slipping; palpate free end

Intercostal Strain: Intercostal muscle strain rib 12 interspace: sharp pain twist/sneeze; palpation intercostal groove tenderness

Manual Therapy: Rib mobilization grade I-IV, soft tissue intercostals, MET inhalation/exhalation, taping support

Breathing Exercises: Diaphragmatic breathing, pursed lip, segmental expansion painful side, avoided Valsalva early

Mobility: Thoracic rotation, serratus wall slide, gentle rib expansion side-lying

Evidence: Rib fracture management: analgesia and pulmonary hygiene (NICE trauma); manual therapy evidence low quality

References: Miller slipping rib syndrome; De Maeseneer costochondritis review; Bordoni intercostal muscles

3. Thoracic Muscles

RECORD: Multifidus (Thoracic)

ID: MUS-T-001

Name: Multifidus (Thoracic)

Origin: Sacrum to C2 spinous processes — thoracic portion from thoracic SPs and laminae

Insertion: Spinous process 2-4 segments superior

Innervation: Medial branch dorsal rami segmental

Blood Supply: Segmental dorsal branches, lumbar/thoracic arteries

Fiber Direction: Oblique cephalad from lateral origin to medial insertion

Primary Function: Segmental spinal stabilization; extension; contralateral rotation

Secondary Function: Controls neutral zone; proprioceptive feedback

Respiratory Function: Minimal direct; stabilizes rib cage base during forced expiration

Stabilizing Role: Deep local stabilizer with transversus thoracis/intercostals

Force Couple: Couples with transversus abdominis via thoracolumbar fascia

EMG: High activation extension and rotation tasks; chronic LBP/thoracic pain shows atrophy

Stretch: Child pose rotation away, seated lumbar lock thoracic stretch indirect

Strengthening: Bird dog, prone extension holds, segmental multifidus cueing at T/L junction

Trigger Points: Paraspinal TP medial to facet line

Pain Referral: Local paraspinal pain; may refer adjacent segment

Clinical Tests: Palpation tone; ultrasound thickness; no isolated strength test

Common Injuries: Strain rare; atrophy with chronic pain; post-surgical dissection

Rehabilitation: Motor control retraining, isometric holds, progressive loading

References: Hides JA multifidus size and LBP; MacDonald multifidus activation

RECORD: Rotatores (Thoracic)

ID: MUS-T-002

Name: Rotatores (Thoracic)

Origin: Thoracic transverse processes

Insertion: Lamina and spinous process 1-2 segments above

Innervation: Medial branch dorsal rami

Blood Supply: Segmental arteries

Fiber Direction: Oblique short fibers — most developed in thoracic spine

Primary Function: Fine segmental rotation and extension

Secondary Function: Proprioception; facet joint positioning

Respiratory Function: Stabilizes during intercostal breathing

Stabilizing Role: Deep segmental stabilizer with multifidus

Force Couple: Works with semispinalis for rotation control

EMG: Moderate in rotation; low in quiet standing

Stretch: Thoracic rotation stretch open book

Strengthening: Prone rotation isometrics, quadruped reach-through

Trigger Points: Deep paraspinal TP difficult to isolate

Pain Referral: Segmental paraspinal ache

Clinical Tests: Segmental rotation resistance
Common Injuries: Overuse with repetitive rotation sports
Rehabilitation: Rotation mobility plus endurance
References: Bogduk deep paraspinal anatomy

RECORD: Semispinalis Thoracis

ID: MUS-T-003
Name: Semispinalis Thoracis
Origin: Transverse processes T6-T10 (variable)
Insertion: Spinous processes T1-T4 and C6-C7 region
Innervation: Dorsal rami thoracic
Blood Supply: Dorsal branches segmental, occipital artery superiorly
Fiber Direction: Long parallel fibers cephalad
Primary Function: Thoracic and cervicothoracic extension; ipsilateral rotation
Secondary Function: Head/neck extension when cervical portion acts
Respiratory Function: Accessory expiration when active
Stabilizing Role: Global extensor stabilizing kyphosis control
Force Couple: Couples with contralateral multifidus
EMG: High in extension against gravity; moderate rotation
Stretch: Prayer stretch, foam roller mid-thoracic extension
Strengthening: Prone Y/T/W, extension endurance holds
Trigger Points: TP along muscle belly paraspinal lateral to SP
Pain Referral: Referral to scapular region and lower cervical
Clinical Tests: Extension strength prone lift
Common Injuries: Strain in weightlifting; trigger point headaches cervicothoracic
Rehabilitation: Gradual extension loading, scapular integration
References: Kapandji spinal muscles; Travell trigger point maps

RECORD: Erector Spinae (Iliocostalis & Longissimus Thoracis)

ID: MUS-T-004
Name: Erector Spinae (Iliocostalis & Longissimus Thoracis)
Origin: Iliac crest, sacrum, lumbar SPs (common erector mass)
Insertion: Ribs (iliocostalis), thoracic TP/SP (longissimus), skull (longissimus capitis cephalad)
Innervation: Dorsal rami segmental
Blood Supply: Cervical, intercostal, lumbar arteries
Fiber Direction: Long vertical columns lateral to transversospinalis
Primary Function: Global spinal extension; lateral flexion; ipsilateral rotation
Secondary Function: Postural maintenance upright stance
Respiratory Function: Forced expiration accessory; stabilizes trunk during cough
Stabilizing Role: Antigravity extensor chain
Force Couple: Force couple with rectus abdominis front-back; with hamstrings via fascia
EMG: High deadlift, extension, heavy carry; moderate walking
Stretch: Child's pose, knee-to-chest partial flexion stretch
Strengthening: Romanian deadlift, back extension, good morning (advanced), row

Trigger Points: Lateral paraspinal TP common
Pain Referral: Diffuse paraspinal referral gluteal (iliocostalis lumborum overlap)
Clinical Tests: Sorensen extension endurance test (lumbar bias but extensor global)
Common Injuries: Muscle strain, DOMS, overuse hyperextension sports
Rehabilitation: Progressive extensor endurance, hip hinge pattern
References: McGill big three adapted; Neumann erector spinae mechanics

RECORD: Serratus Posterior Superior

ID: MUS-T-005
Name: Serratus Posterior Superior
Origin: Ligamentum nuchae, C7-T3 SPs
Insertion: Ribs 2-5 superior aspect
Innervation: Intercostal nerves T1-T5 anterior rami branches
Blood Supply: Intercostal and supreme intercostal arteries
Fiber Direction: Oblique inferolateral
Primary Function: Elevates ribs 2-5 (accessory inspiration)
Secondary Function: May extend spine slightly via rib attachment
Respiratory Function: Classic accessory inspiratory muscle — disputed by some modern EMG studies
Stabilizing Role: Stabilizes upper rib cage during arm elevation
Force Couple: Couples with scalenes and upper trapezius during apical breathing
EMG: Low-moderate forced inspiration; minimal quiet breathing
Stretch: Side-lying rib depression stretch
Strengthening: Deep breathing resisted, serratus anterior strengthening primary clinically
Trigger Points: TP deep to rhomboids
Pain Referral: Deep ache under scapula, referral to posterior shoulder
Clinical Tests: Palpation deep rhomboid layer; breathing repro
Common Injuries: Overuse with apical breathing pattern dysfunction
Rehabilitation: Diaphragmatic retraining, scapular exercises
References: Bordoni serratus posterior nomenclature debate

RECORD: Serratus Posterior Inferior

ID: MUS-T-006
Name: Serratus Posterior Inferior
Origin: T11-L2 spinous processes, thoracolumbar fascia
Insertion: Ribs 9-12 inferior border
Innervation: Intercostal nerves T9-L2
Blood Supply: Intercostal and lumbar arteries
Fiber Direction: Oblique inferolateral
Primary Function: Depresses lower ribs (accessory expiration)
Secondary Function: Assists trunk extension via rib pull
Respiratory Function: Forced expiration and coughing
Stabilizing Role: Lower rib cage stabilization
Force Couple: Couples with obliques and QL
EMG: Moderate forced expiration

Stretch: Side-lying lower rib expansion
Strengthening: Oblique crunches, expiration against resistance
Trigger Points: TP over lower ribs posterior
Pain Referral: Lower thoracic flank pain
Clinical Tests: Palpation inferior rib angle tenderness
Common Injuries: Strain with persistent cough
Rehabilitation: Breathing retraining post-respiratory illness
References: Moore Clinically Oriented Anatomy

RECORD: External Intercostals

ID: MUS-T-007
Name: External Intercostals
Origin: Inferior border rib above
Insertion: Superior border rib below — oblique inferoanterior
Innervation: Intercostal nerve same level
Blood Supply: Intercostal artery
Fiber Direction: Oblique anteroinferior (same direction as external oblique)
Primary Function: Elevate ribs in inspiration; stiffen interspace
Secondary Function: Prevent inward sucking during inspiration
Respiratory Function: Primary inspiratory intercostal layer
Stabilizing Role: Rib cage hoop integrity
Force Couple: Couples with diaphragm descent
EMG: Active quiet and forced inspiration
Stretch: Side-lying open painful side, rotation away
Strengthening: Inspiratory resisted breathing, swimming endurance
Trigger Points: TP in intercostal groove
Pain Referral: Anterior chest wall along rib
Clinical Tests: Palpation intercostal groove; pain with deep breath
Common Injuries: Intercostal strain, tear rare
Rehabilitation: Gradual return to rotation sports
References: De Troyer respiratory mechanics

RECORD: Internal Intercostals

ID: MUS-T-008
Name: Internal Intercostals
Origin: Superior border costal groove rib below
Insertion: Inferior border rib above — oblique inferoposterior
Innervation: Intercostal nerve
Blood Supply: Intercostal vessels
Fiber Direction: Oblique posteroinferior (perpendicular to external)
Primary Function: Depress ribs (esp lower interspaces expiration); subcostales portion
Secondary Function: Stabilize interspace
Respiratory Function: Forced expiration primary
Stabilizing Role: Prevent excessive rib elevation

Force Couple: Couples with transversus thoracis
EMG: High forced expiration cough
Stretch: Seated slump opening anterior chest
Strengthening: Forced expiration resisted, pursed lip
Trigger Points: Deep TP near sternum posterior intercostal
Pain Referral: Anterior chest referral
Clinical Tests: Pain with cough/manual intercostal resistance
Common Injuries: Intercostal strain common
Rehabilitation: Breathing control, gradual loading
References: Bordoni intercostal functional anatomy

RECORD: Innermost Intercostals (Transversus Thoracis component)

ID: MUS-T-009
Name: Innermost Intercostals (Transversus Thoracis component)
Origin: Inner surface thoracic cage — transversus thoracis from sternum; innermost intercostals in interspaces
Insertion: Opposite rib/costal cartilage internal surface
Innervation: Intercostal nerves
Blood Supply: Intercostal vessels
Fiber Direction: Transverse horizontal on inner wall
Primary Function: Depress ribs; stiffen thoracic wall
Secondary Function: Protect viscera; minimal ROM change
Respiratory Function: Expiration; stabilize during Valsalva
Stabilizing Role: Deep thoracic wall stabilizer
Force Couple: Works with internal intercostals and diaphragm
EMG: Moderate Valsalva/cough
Stretch: Diaphragmatic expansion indirectly
Strengthening: Transversus thoracis isometric cough training
Trigger Points: Not commonly described TP
Pain Referral: Deep sternal pain rare
Clinical Tests: Not isolated clinically
Common Injuries: Rare
Rehabilitation: Post-operative cardiac/thoracic surgery breathing
References: Standring Gray's Anatomy thoracic wall

RECORD: Diaphragm

ID: MUS-T-010
Name: Diaphragm
Origin: Xiphoid, lower 6 ribs costal margin, L1-L3 vertebral bodies/right crus
Insertion: Central tendon
Innervation: Phrenic nerve C3-C5
Blood Supply: Pericardiophrenic, musculophrenic, inferior phrenic arteries
Fiber Direction: Radial fibers to central tendon; crural portions
Primary Function: Primary inspiratory muscle — 70-80% tidal volume quiet breathing
Secondary Function: Increases intra-abdominal pressure; sphincter esophagus crural part

Respiratory Function: Essential respiration; accessory roles postural

Stabilizing Role: Lumbopelvic stability via IAP with pelvic floor/transversus

Force Couple: Couples with pelvic floor and TA — anticipatory postural

EMG: Continuous tonic activity; increases with exercise

Stretch: Crocodile breathing, child pose side stretch indirect

Strengthening: Supine 90-90 breathing, weighted breathing, dead bug

Trigger Points: Trigger points crura and costal margin (controversial)

Pain Referral: Referred shoulder (Kehr phenomenon surgical); neck via scalene compensation

Clinical Tests: Observation breathing pattern; hi-lo test; MIP/MEP spirometry medical

Common Injuries: Hiatus hernia, eventration, paralysis phrenic nerve

Rehabilitation: Diaphragmatic breathing retraining, post-COVID pulmonary rehab

References: Kolar et al postural-respiratory function; McKenzie breathing

RECORD: Latissimus Dorsi

ID: MUS-T-011

Name: Latissimus Dorsi

Origin: Spinous processes T7-S5, thoracolumbar fascia, iliac crest, lower 3-4 ribs

Insertion: Floor of bicipital groove humerus (teres major junction)

Innervation: Thoracodorsal nerve C6-C8

Blood Supply: Thoracodorsal artery

Fiber Direction: Broad fan — horizontal lower, oblique superior fibers

Primary Function: Shoulder extension, adduction, internal rotation

Secondary Function: Depression of scapula via humeral action; forced expiration accessory

Respiratory Function: Accessory expiration when arms fixed

Stabilizing Role: Thoracolumbar fascia tension; anti-flexion

Force Couple: Couples with glute max in posterior chain; contralateral lat-glute

EMG: High pull-ups, swimming, throwing deceleration

Stretch: Child pose arm reach, overhead lat stretch

Strengthening: Pull-up, lat pulldown, single arm row, straight arm pulldown

Trigger Points: TP inferior angle scapula region

Pain Referral: Referral to mid-back and down arm posterior

Clinical Tests: Reach test; lat length overhead

Common Injuries: Strain, tendinopathy humeral insertion, overuse swimmers

Rehabilitation: Gradual pull progression, scapular control

References: Neumann latissimus mechanics; Escamilla EMG lat pull

RECORD: Rhomboid Major

ID: MUS-T-012

Name: Rhomboid Major

Origin: Spinous processes T2-T5

Insertion: Medial border scapula below spine

Innervation: Dorsal scapular nerve C4-C5

Blood Supply: Dorsal scapular artery

Fiber Direction: Oblique inferolateral

Primary Function: Scapular retraction; downward rotation component; elevation assist

Secondary Function: Scapular stabilization during rowing

Respiratory Function: Minimal respiratory

Stabilizing Role: Maintains scapula on thorax

Force Couple: Couples with middle trapezius retraction; antagonist serratus anterior

EMG: High rowing, high retraction holds

Stretch: Wall angel, doorway stretch pec minor indirect

Strengthening: Row, face pull, prone rhomboid squeeze

Trigger Points: TP medial scapular border

Pain Referral: Interscapular ache, referral to rhomboid region

Clinical Tests: Manual muscle test retraction; scapular dyskinesis observation

Common Injuries: Overuse rowing; strain

Rehabilitation: Scapular retraction endurance, lower trap balance

References: Cools AM scapular muscle EMG

RECORD: Rhomboid Minor

ID: MUS-T-013

Name: Rhomboid Minor

Origin: Spinous processes C7-T1

Insertion: Medial scapular border at spine level

Innervation: Dorsal scapular nerve

Blood Supply: Dorsal scapular artery

Fiber Direction: Oblique inferolateral short fibers

Primary Function: Retraction; slight elevation scapula

Secondary Function: Cervicothoracic transition stability

Respiratory Function: Minimal

Stabilizing Role: Upper scapular control

Force Couple: With levator scapulae for elevation-retraction

EMG: Moderate retraction tasks

Stretch: Seated rotation stretch

Strengthening: Same as rhomboid major

Trigger Points: TP at medial superior scapula

Pain Referral: Upper interscapular pain, neck referral

Clinical Tests: Palpation C7-T1 region

Common Injuries: Postural overload desk workers

Rehabilitation: Postural education, thoracic mobility

References: Travell rhomboid trigger points

RECORD: Trapezius — Upper Fibers

ID: MUS-T-014

Name: Trapezius — Upper Fibers

Origin: Occipital protuberance, nuchal ligament, C7 SP

Insertion: Lateral third clavicle, acromion

Innervation: Accessory nerve XI, C3-C4 proprioception

Blood Supply: Transverse cervical/suprascapular arteries
Fiber Direction: Oblique inferolateral
Primary Function: Scapular elevation; upward rotation assist; neck extension
Secondary Function: Occipital stabilization
Respiratory Function: Apical breathing accessory when hypertonic
Stabilizing Role: Cervicothoracic posture
Force Couple: Couples with serratus anterior for upward rotation
EMG: High overhead lifting, shrugs
Stretch: Upper trap stretch side bend away
Strengthening: Shrug, overhead carry (moderate load)
Trigger Points: TP common at angle neck-shoulder
Pain Referral: Headache, neck pain, lateral shoulder
Clinical Tests: Shrug strength; upper trap overactivity in impingement
Common Injuries: Tension neck syndrome, whiplash
Rehabilitation: Scapular motor control, lower trap emphasis
References: Cools trapezius EMG divisions

RECORD: Trapezius — Middle Fibers

ID: MUS-T-015
Name: Trapezius — Middle Fibers
Origin: C7-T3 spinous processes
Insertion: Spine of scapula
Innervation: Accessory nerve XI
Blood Supply: Transverse cervical artery
Fiber Direction: Horizontal
Primary Function: Scapular retraction
Secondary Function: Thoracic extension assist via scapula
Respiratory Function: Minimal
Stabilizing Role: Scapular centralization
Force Couple: Force couple with serratus anterior protraction/retraction balance
EMG: High rowing horizontal abduction
Stretch: Pec stretch, thoracic extension
Strengthening: Row, band pull-apart, prone T
Trigger Points: TP interscapular
Pain Referral: Interscapular pain
Clinical Tests: Retraction MMT
Common Injuries: Postural strain
Rehabilitation: Retraction endurance
References: Reinold scapular biomechanics

RECORD: Trapezius — Lower Fibers

ID: MUS-T-016
Name: Trapezius — Lower Fibers
Origin: T4-T12 spinous processes

Insertion: Medial spine scapula apex tubercle region
Innervation: Accessory nerve XI
Blood Supply: Transverse cervical artery
Fiber Direction: Oblique superolateral
Primary Function: Scapular depression; upward rotation; posterior tilt assist
Secondary Function: Counteracts upper trap elevation dominance
Respiratory Function: Minimal
Stabilizing Role: Scapulothoracic rhythm upward rotation
Force Couple: Primary upward rotation couple with serratus anterior
EMG: High overhead, swimming recovery phase
Stretch: Child pose arms overhead
Strengthening: Prone Y, wall slide lower trap, scaption
Trigger Points: TP inferior medial scapula
Pain Referral: Interscapular lower pain
Clinical Tests: Prone Y hold endurance test
Common Injuries: Weakness in overhead athletes impingement
Rehabilitation: Lower trap strengthening progression
References: Kibler scapula dyskinesis

4. Breathing Mechanics

RECORD: Diaphragmatic (Abdominal) Breathing

ID: BRTH-001

Movement: Diaphragmatic (Abdominal) Breathing

Muscles: Diaphragm primary; external intercostals; minimal accessory

Joint Motion: Costovertebral and sternocostal glides; thoracic spine subtle extension on deep inhale

Rib Motion: Lower rib lateral expansion (bucket/caliper); minimal superior rib rise

Thoracic Motion: Lower thoracic expansion; posterior-lateral excursion

Accessory Muscles: Scalenes, SCM, upper trap minimally active

Diaphragm Function: Descent increases vertical thoracic dimension; dome flattens; efficient O₂ cost

Pressure Changes: Inspiratory negative intrathoracic; increased intra-abdominal pressure coordinated with pelvic floor

Clinical Assessment: Observe hand on abdomen vs chest; hi-lo test supine; count respiratory rate

Breathing Pattern: Slow nasal inhale 360-degree rib expansion; exhale via pursed lips or relaxed mouth

Common Dysfunction: Paradoxical pattern, hyperventilation, stress breathing upper chest only

Corrective Exercises: 90-90 breathing, crocodile breathing, balloon blowing exhalation, dead bug coordination

Evidence: Diaphragmatic training improves GERD, anxiety symptoms Level B; athletic performance mixed

References: McConnell respiratory; Chaitow breathing pattern disorders

RECORD: Apical (Clavicular) Breathing

ID: BRTH-002

Movement: Apical (Clavicular) Breathing

Muscles: Scalenes, SCM, upper trapezius, pectoralis minor, levator scapulae

Joint Motion: First rib elevation; limited costovertebral bucket handle

Rib Motion: Superior rib elevation pump handle dominant; poor lateral expansion

Thoracic Motion: Upper thoracic extension; cervical protraction often coupled

Accessory Muscles: High accessory muscle reliance

Diaphragm Function: Diaphragm underutilized; shallow tidal volume

Pressure Changes: Rapid shallow breathing; increased accessory muscle oxygen demand

Clinical Assessment: Observe superior rib cage rise; palpate SCM/scalene overactivity

Breathing Pattern: Visible shoulder elevation on inhale; mouth breathing common

Common Dysfunction: TOS risk, neck pain, hyperventilation syndrome, poor CO₂ tolerance

Corrective Exercises: Diaphragm retraining, soft tissue scalenes, first rib mobilization, postural correction

Evidence: Linked to chronic neck pain; causation vs correlation debated

References: Bradley apical breathing and neck disorders

RECORD: Accessory Breathing

ID: BRTH-003

Movement: Accessory Breathing

Muscles: SCM, scalenes, pec minor, serratus anterior, upper trap, erector spinae, lat (fixed arms)

Joint Motion: Increased costovertebral motion; scapular protraction/elevation

Rib Motion: Exaggerated pump and bucket handle all levels

Thoracic Motion: Extension and rotation with forced breath

Accessory Muscles: All listed accessory muscles — active in exercise and respiratory distress

Diaphragm Function: Diaphragm plus accessory for increased ventilation demand

Pressure Changes: Large negative intrapleural pressure; IAP rise with effort

Clinical Assessment: Observation at rest vs exertion; capnography if hyperventilation

Breathing Pattern: Normal during high-intensity exercise; pathological at rest

Common Dysfunction: Chronic overuse accessory at rest; muscle fatigue; poor recovery

Corrective Exercises: Graded exercise tolerance, diaphragm priority at rest, interval training

Evidence: Accessory muscle fatigue in COPD well documented; athletic extrapolation

References: De Troyer accessory muscle respiratory mechanics

RECORD: Paradoxical Breathing

ID: BRTH-004

Movement: Paradoxical Breathing

Muscles: Diaphragm paralysis/dysfunction; abdominal wall paradoxical inward draw

Joint Motion: Costovertebral motion reversed or absent; sternal retraction on inhale

Rib Motion: Ribs may draw inward on inspiration instead of expansion

Thoracic Motion: Flaring or collapse inconsistent with normal kinematics

Accessory Muscles: Compensatory extreme accessory use

Diaphragm Function: Absent or weak diaphragm contraction unilateral/bilateral

Pressure Changes: Inefficient ventilation; may see indrawing intercostal spaces

Clinical Assessment: Visual inspection supine; fluoroscopy/ultrasound medical; MIP testing

Breathing Pattern: Inward abdominal wall motion during inhale

Common Dysfunction: Phrenic nerve injury, COPD hyperinflation, post-surgical, obesity

Corrective Exercises: Medical workup first; inspiratory muscle training if appropriate; surgical plication severe cases

Evidence: Documented in diaphragm paralysis; rehab case series inspiratory muscle training

References: McCool diaphragm dysfunction review

5. Scapular Biomechanics

RECORD: Elevation

ID: SCAP-001

Movement: Elevation

Plane: Frontal

Axis: Anterior-posterior through scapula

Primary Muscles: Upper trapezius, levator scapulae

Secondary Muscles: Rhomboids minor assist

Force Couples: Upper trap/levator vs lower trap depression couple

Joint Contribution: AC joint slight; scapulothoracic glide superior

Scapulohumeral Rhythm: Necessary first 30 deg humeral elevation before upward rotation dominates

EMG: High shrugs; moderate initial abduction

Clinical Importance: Overhead reach, carrying loads, apical breathing

Common Dysfunctions: Upper trap dominance, levator tightness, TOS posture

Assessment: Shrug test; observation overhead

Corrective Exercises: Lower trap emphasis, upper trap stretch, scapular setting

Progressions: Wall slide without elevation → weighted overhead

References: Kibler scapula kinematics

RECORD: Depression

ID: SCAP-002

Movement: Depression

Plane: Frontal

Axis: AP axis

Primary Muscles: Lower trapezius, pectoralis minor (from rib pull), latissimus via humerus fixed

Secondary Muscles: Serratus anterior lower fibers

Force Couples: Lower trap/serratus vs upper trap

Joint Contribution: Scapulothoracic inferior glide

Scapulohumeral Rhythm: Counterpart to elevation in arm lowering

EMG: High lowering phase pull-up; rowing eccentric

Clinical Importance: Pull-ups, dips, push-up plus

Common Dysfunctions: Depression weakness in overhead athletes

Assessment: Prone Y; observation sag during push-up

Corrective Exercises: Push-up plus, dip support holds

Progressions: Band assisted → full dip depression control

References: Reinold scapular depression

RECORD: Upward Rotation

ID: SCAP-003

Movement: Upward Rotation

Plane: Coronal oblique

Axis: Transverse through scapula

Primary Muscles: Serratus anterior, lower trapezius

Secondary Muscles: Upper trapezius

Force Couples: Serratus + lower trap primary couple — most critical for overhead

Joint Contribution: GH abduction/flexion coupled 2:1 rhythm overall

Scapulohumeral Rhythm: ~1 deg scap upward rotation per 2 deg humeral elevation after initial phase

EMG: Very high overhead activities swimming throwing

Clinical Importance: Impingement prevention; full flexion 180 deg

Common Dysfunctions: Serratus weakness winging; lower trap insufficiency

Assessment: Scapular dyskinesis type II; wall slide; LSST

Corrective Exercises: Serratus punch, wall slide, prone Y

Progressions: Isometric → band → full overhead sport

References: Ludewig scapulothoracic rhythm

RECORD: Downward Rotation

ID: SCAP-004

Movement: Downward Rotation

Plane: Coronal oblique

Axis: Transverse

Primary Muscles: Rhomboids major/minor, levator scapulae, pec minor

Secondary Muscles: Middle trapezius eccentric

Force Couples: Rhomboids/levator vs serratus/lower trap

Joint Contribution: Scapulothoracic downward rotation; glenoid tilts inferiorly

Scapulohumeral Rhythm: Coupled with humeral extension/adduction during pull-through

EMG: High rowing, lat pull, climbing pull phases

Clinical Importance: Climbing, rowing, swimming pull-through, cross-country ski poling

Common Dysfunctions: Overactive rhomboids postural pain; scapular downward rotation dominance

Assessment: Row end-range retraction hold; observation during lat pull

Corrective Exercises: Row, face pull with external rotation, rhomboid endurance

Progressions: Light band → heavy row → weighted pull-ups

References: Cools rowing EMG

RECORD: Protraction

ID: SCAP-005

Movement: Protraction

Plane: Transverse

Axis: Vertical sagittal

Primary Muscles: Serratus anterior

Secondary Muscles: Pectoralis major/minor assist

Force Couples: Serratus vs rhomboids retraction couple

Joint Contribution: Scapula slides around rib cage anteriorly

Scapulohumeral Rhythm: Reach and punch motions; serratus-on-wall glide

EMG: High punching, push-up plus, throwing follow-through

Clinical Importance: Push reach sports, bench press, overhead press stability

Common Dysfunctions: Serratus weakness winging; long thoracic nerve injury

Assessment: Push-up plus test; wall punch; dynamic scapular dyskinesis

Corrective Exercises: Serratus wall slide, push-up plus, band protraction

Progressions: Knees → standard → single arm → plyometric punch

References: Martin serratus anterior review

RECORD: Retraction

ID: SCAP-006

Movement: Retraction

Plane: Transverse

Axis: Vertical sagittal

Primary Muscles: Middle trapezius, rhomboids

Secondary Muscles: Lat posterior fibers

Force Couples: Middle trap/rhomboid vs serratus/pec minor

Joint Contribution: Scapula medial glide on thorax

Scapulohumeral Rhythm: Rowing phase; posture

EMG: High horizontal pull

Clinical Importance: Desk posture dysfunction; interscapular pain

Common Dysfunctions: Rounded shoulders; poor scapular control

Assessment: Retraction endurance test

Corrective Exercises: Band pull-apart, row, prone T

Progressions: Endurance → load

References: Cools retraction EMG

RECORD: Internal Rotation (Scapular)

ID: SCAP-007

Movement: Internal Rotation (Scapular)

Plane: Transverse

Axis: Vertical

Primary Muscles: Pectoralis minor, levator scapulae, upper trap

Secondary Muscles: Anterior deltoid via humerus

Force Couples: Pec minor vs lower trap posterior tilt

Joint Contribution: Coracoid moves inferior-medial

Scapulohumeral Rhythm: Often coupled protraction

EMG: Moderate bench press bottom

Clinical Importance: Forward head posture component

Common Dysfunctions: Pec minor shortness; thoracic outlet risk

Assessment: Horizon test pec minor length

Corrective Exercises: Pec stretch, lower trap/Y raises

Progressions: Stretch → strengthen posterior chain

References: Borstad pec minor length

RECORD: External Rotation (Scapular)

ID: SCAP-008

Movement: External Rotation (Scapular)

Plane: Transverse

Axis: Vertical

Primary Muscles: Middle/lower trapezius, serratus

Secondary Muscles: Rhomboids eccentric control

Force Couples: Posterior tilt couple with external rotation

Joint Contribution: Posterior scapula rotates on rib cage

Scapulohumeral Rhythm: Overhead external rotation preparation

EMG: High throwing cocking phase indirectly

Clinical Importance: Throwing athletes scapular control

Common Dysfunctions: Lost ER scapular in impingement

Assessment: Observation throwing; wall test

Corrective Exercises: External rotation rows, prone horizontal abduction ER

Progressions: Band ER → weighted

References: Myers scapular ER throwing

RECORD: Anterior Tilt

ID: SCAP-009

Movement: Anterior Tilt

Plane: Sagittal

Axis: Medial-lateral scapula

Primary Muscles: Pectoralis minor, levator scapulae

Secondary Muscles: Short head biceps via coracoid

Force Couples: Pec minor vs lower trap posterior tilt

Joint Contribution: Inferior angle lifts off thorax

Scapulohumeral Rhythm: Early elevation; poor overhead

EMG: Moderate unless pathological

Clinical Importance: Impingement; subacromial space reduction

Common Dysfunctions: Excessive anterior tilt dyskinesis

Assessment: Inferior angle palpation off rib cage

Corrective Exercises: Lower trap, serratus, thoracic extension

Progressions: Wall angels, foam roller extension

References: Ludewig anterior tilt impingement

RECORD: Posterior Tilt

ID: SCAP-010

Movement: Posterior Tilt

Plane: Sagittal

Axis: Medial-lateral

Primary Muscles: Lower trapezius, serratus anterior

Secondary Muscles: Middle trapezius

Force Couples: Lower trap/serratus vs pec minor

Joint Contribution: Superior scapula rotates posterior; increases subacromial space

Scapulohumeral Rhythm: Critical 120-180 deg elevation

EMG: Very high overhead throwing serving

Clinical Importance: Impingement prevention

Common Dysfunctions: Insufficient posterior tilt in overhead athletes

Assessment: Wall slide observation; Kibler dyskinesis

Corrective Exercises: Prone Y, wall slide, scaption

Progressions: Low load → sport specific

References: Reinold posterior tilt exercises

6. Thoracic Outlet Syndrome

RECORD: TOS — Neurogenic (nTOS)

ID: TOS-001

Type: Neurogenic TOS (most common ~95%)

Compression Site: Interscalene triangle, costoclavicular space, subpectoral/minor space

Structures Compressed: Brachial plexus lower trunk (C8-T1) most often; upper trunk less common

Symptoms: Unilateral arm paresthesia ulnar distribution, hand intrinsic weakness, neck-shoulder pain, worse overhead

Differential Diagnosis: Cervical radiculopathy, peripheral neuropathy, CTS, Pancoast tumor, MS

Special Tests: Roos, Adson, Wright, upper limb tension, Tinel at scalene

Imaging: X-ray cervical rib; MRI brachial plexus; EMG/NCS if chronic

Conservative Management: Postural correction, first rib mobilization, scalene soft tissue, strengthening middle/lower trap

Exercises: Chin tuck, scapular retraction, nerve glides cautious, diaphragmatic breathing

Return to Sport: Overhead tolerance without paresthesia; strength symmetric; gradual throw/swim progression

Evidence: Conservative success 50-90% mild-moderate; surgery reserved failed rehab documented vascular or severe neuro

References: Sanders TOS classification; Nord KM conservative nTOS outcomes

RECORD: TOS — Venous (vTOS)

ID: TOS-002

Type: Venous TOS (Paget-Schroetter effort thrombosis)

Compression Site: Costoclavicular space subclavian vein

Structures Compressed: Subclavian vein

Symptoms: Arm swelling, cyanosis, heaviness, venous distention, sudden after exertion

Differential Diagnosis: DVT other causes, lymphedema, cellulitis

Special Tests: Adson less specific; imaging primary — duplex US, venography

Imaging: Duplex ultrasound, CT/MR venography, D-dimer

Conservative Management: Anticoagulation medical; thrombolysis acute; then rib resection often surgical

Exercises: Post-surgical rehab only after medical clearance; scapular stabilization later

Return to Sport: Medical team clearance; often months; contact sports individualized

Evidence: Acute thrombolysis plus decompression standard; delay increases chronic disability

References: Illig venous TOS guidelines; Thompson effort thrombosis

RECORD: TOS — Arterial (aTOS)

ID: TOS-003

Type: Arterial TOS (rare)

Compression Site: Subclavian artery scalene/costoclavicular

Structures Compressed: Subclavian artery

Symptoms: Arm claudication, cold pale hand, pulse deficit, embolic phenomena

Differential Diagnosis: Peripheral arterial disease, embolic stroke young patient, vasculitis

Special Tests: Adson pulse change; imaging mandatory

Imaging: CT/MR angiography, arterial duplex

Conservative Management: Rarely sufficient; surgical decompression/resection cervical rib

Exercises: Post-op rehab similar neurogenic without vascular stress

Return to Sport: Specialist guided; vascular monitoring

Evidence: Surgery recommended to prevent embolic complications

References: Illig arterial TOS; Sanders aTOS outcomes

RECORD: TOS Special Test — Roos (Elevated Arm Stress Test)

ID: TOS-T-001

Type: Special Test — Roos

Compression Site: Costoclavicular and subpectoral spaces under load

Structures Compressed: Brachial plexus and subclavian vessels

Symptoms: Reproduction paresthesia or heaviness within 3 minutes

Differential Diagnosis: Not diagnostic alone — combine history and other tests

Special Tests: Procedure: arms 90 deg abduction ER, open/close hands 3 min; positive = symptom reproduction

Imaging: N/A for test

Conservative Management: N/A

Exercises: N/A

Return to Sport: N/A

Evidence: Sensitivity moderate; specificity low — better as stress reproducer than rule-out

References: Gillen TOS tests systematic review

RECORD: TOS Special Test — Adson (Scalene Maneuver)

ID: TOS-T-002

Type: Special Test — Adson

Compression Site: Interscalene triangle

Structures Compressed: Subclavian artery (primarily); brachial plexus

Symptoms: Pulse diminution and/or symptom reproduction

Differential Diagnosis: Vascular vs neurogenic differentiation

Special Tests: Extend neck ipsilateral, rotate to affected side, hold breath; palpate radial pulse

Imaging: Follow positive with duplex if vascular suspected

Conservative Management: N/A

Exercises: N/A

Return to Sport: N/A

Evidence: Low sensitivity/specificity for nTOS; some utility vascular

References: Rayan Adson test critique

RECORD: TOS Special Test — Wright (Hyperabduction)

ID: TOS-T-003

Type: Special Test — Wright

Compression Site: Subpectoral/minor space, costoclavicular

Structures Compressed: Neurovascular bundle under pectoralis minor

Symptoms: Paresthesia or pulse change with hyperabduction

Differential Diagnosis: Pec minor syndrome overlap

Special Tests: Arm passively hyperabducted 180 deg; monitor pulse and symptoms 1-3 min

Imaging: N/A

Conservative Management: N/A

Exercises: N/A

Return to Sport: N/A

Evidence: Moderate sensitivity neurogenic in some series

References: Gillen TOS physical exam

RECORD: TOS Special Test — Costoclavicular (Military Brace)

ID: TOS-T-004

Type: Special Test — Costoclavicular

Compression Site: Between clavicle and first rib

Structures Compressed: Subclavian artery/vein, brachial plexus

Symptoms: Pulse reduction, arm symptoms

Differential Diagnosis: First rib fixation, clavicle malunion

Special Tests: Shoulders thrust back and down (military posture); hold breath optional

Imaging: X-ray if bony anomaly

Conservative Management: N/A

Exercises: N/A

Return to Sport: N/A

Evidence: Part of TOS test battery; not standalone diagnostic

References: Sanders physical diagnosis TOS

7. Myofascial System

RECORD: Thoracolumbar Fascia (TLF)

ID: FAS-T-001

Structure: Thoracolumbar Fascia (TLF)

Region: Posterior thorax and lumbar region

Continuity: Continuous with latissimus, glute max, serratus posterior, internal oblique/aponeurosis

Biomechanical Role: Transfers load between upper and lower extremity; IAP containment; extensor force transmission

Force Transmission: Posterior chain tension; contralateral lat-glute coupling in gait

Sliding Mechanism: Multi-layer sliding thoracolumbar junction; hyaluronan-dependent

Clinical Importance: Central to LBP and thoracic stiffness; lifting mechanics

Restrictions: Thickening/adhesion post-injury; connective tissue densification

Pain Referral: Diffuse back/flank pain; referred gluteal

Assessment: Palpation shear layers; movement loss trunk rotation

Treatment: Myofascial release, instrument assisted soft tissue, movement hydration, loading

Evidence: Stecco fascial manipulation RCTs emerging; McGill posterior chain

References: Schleip fascia as sensory organ; Willard TLF anatomy

RECORD: Deep Cervical Fascia

ID: FAS-T-002

Structure: Deep Cervical Fascia

Region: Invests neck muscles; pretracheal, prevertebral layers extending to upper thoracic

Continuity: Continuous with thoracic prevertebral fascia to mediastinum

Biomechanical Role: Compartmentalization; carotid sheath; support viscera and deep neck flexors

Force Transmission: Cervicothoracic force to first rib and upper thoracic spine

Sliding Mechanism: Sliding between SCM and deep layers; important post-whiplash

Clinical Importance: TOS, cervicogenic headache, upper thoracic pain

Restrictions: Post-traumatic restriction; forward head posture

Pain Referral: Occipital and upper thoracic referral

Assessment: Cervical rotation coupled with upper thoracic assessment

Treatment: Deep neck flexor training, soft tissue, first rib mobilization

Evidence: Limited high-quality RCTs; clinical integration common

References: Standring Gray's fascia; Jull deep neck flexor

RECORD: Brachial Fascia (Axillary/Clavipectoral continuity)

ID: FAS-T-003

Structure: Brachial Fascia (Axillary/Clavipectoral continuity)

Region: Shoulder girdle extending from clavipectoral to arm

Continuity: Clavipectoral fascia continuous with pectoralis minor, axillary fascia, serratus

Biomechanical Role: Encloses neurovascular bundle; scapulothoracic glide interface

Force Transmission: Load transfer push/pull to thoracic wall

Sliding Mechanism: Sliding scapula on thorax beneath brachial fascia

Clinical Importance: TOS, pec minor syndrome, scapular dyskinesis

Restrictions: Pec minor/adhesive restriction overhead
Pain Referral: Anterior chest, medial arm referral
Assessment: Overhead reach, pec minor length, TOS tests
Treatment: Fascial release pec minor, serratus strengthening
Evidence: Anatomy strong; treatment evidence case-based
References: Moore clavipectoral fascia; Stecco upper limb

RECORD: Clavipectoral Fascia

ID: FAS-T-004

Structure: Clavipectoral Fascia

Region: Between clavicle and pectoralis minor; costocoracoid ligament region

Continuity: Blends with costoclavicular membrane; suspensory ligaments

Biomechanical Role: Suspends subclavian vessels and brachial plexus; divides axillary compartments

Force Transmission: Compression site TOS costoclavicular

Sliding Mechanism: Limited sliding — dense fascial ring

Clinical Importance: Critical TOS landmark; post-surgical scarring

Restrictions: Post-op TOS, clavicle fracture malunion

Pain Referral: Medial arm, paresthesia TOS pattern

Assessment: Pulse exam Adson; palpation infraclavicular

Treatment: Conservative postural; surgical if vascular

Evidence: Illig TOS surgical anatomy

References: Illig; Sanders costoclavicular space

RECORD: Palmar Fascia (Connection via SBL)

ID: FAS-T-005

Structure: Palmar Fascia (Connection via SBL)

Region: Hand — linked via Superficial Back Line to thoracolumbar fascia

Continuity: Continuous myofascial chain through epicondyle, triceps, lat, TLF (Anatomy Trains model)

Biomechanical Role: Grip force transmission to axial skeleton in climbing/lifting

Force Transmission: Long chain tension — elbow/wrist/thoracic coupling

Sliding Mechanism: Sliding carpal tunnel retinaculum separate; fascial continuity functional

Clinical Importance: Dupuytren adjacent; climber flexor chain overload affects posture

Restrictions: Flexor retinaculum thickening; chain tightness

Pain Referral: Medial epicondyle, thoracic extension loss secondary

Assessment: Grip strength, thoracic rotation screen

Treatment: Chain stretching, thoracic mobility, grip training balanced

Evidence: Anatomy Trains model partially validated; Dupuytren surgical evidence strong

References: Myers Anatomy Trains; Wilbrand Dupuytren

8. Myofascial Chains

RECORD: Superficial Back Line (SBL)

ID: CHN-T-001

Chain: Superficial Back Line (SBL)

Muscles Included: Plantar fascia, gastroc-soleus, hamstrings, sacrotuberous, erector spinae, splenius capitis, scalp fascia

Function: Extends body; maintains upright posture; limits forward bend

Clinical Importance: Flexibility deficits limit toe touch and thoracic flexion; linked to posterior chain tightness

Sports: Running, deadlift, rowing, swimming starts

Assessment: Toe touch, squat depth, thoracic flexion seated

Exercises: Downward dog, hamstring stretch, thoracic cat, foam roll erectors

Evidence: Myers model widely used clinically; partial EMG/kinetic validation

References: Myers Anatomy Trains 4th ed

RECORD: Superficial Front Line (SFL)

ID: CHN-T-002

Chain: Superficial Front Line (SFL)

Muscles Included: Tibialis anterior, quads, rectus abdominis, sternalis/pectoralis, SCM

Function: Flexes trunk and hip; balances SBL

Clinical Importance: Dominance in sitting posture; pec tightness thoracic kyphosis

Sports: Sprinting start, kicking, gymnastics

Assessment: Thomas test hip, pec minor length, thoracic extension

Exercises: Hip flexor stretch, thoracic extension mobilization, chin tuck

Evidence: Force balance front-back clinically applied

References: Myers; Liebensohn Janda crossed syndromes overlap

RECORD: Deep Front Line (DFL)

ID: CHN-T-003

Chain: Deep Front Line (DFL)

Muscles Included: Tibialis posterior, popliteus, adductors, psoas, diaphragm, TFL/deep lateral pelvis, scalenes

Function: Core stability; breathing; axial support

Clinical Importance: Breathing-posture integration; psoas-diaphragm coupling

Sports: All sports requiring trunk stability — gymnastics, combat, throwing

Assessment: Hi-lo breathing, psoas length, hip internal rotation

Exercises: 90-90 breathing, psoas release cautious, dead bug, plank variations

Evidence: Kolar DNS overlaps; growing fascial research

References: Myers; Kolar Dynamic Neuromuscular Stabilization

RECORD: Spiral Line (SL)

ID: CHN-T-004

Chain: Spiral Line (SL)

Muscles Included: Splenius capitis to opposite rhomboids, serratus anterior, external oblique, ITB, tibialis anterior

Function: Rotational force transmission across body

Clinical Importance: Thoracic rotation restrictions affect gait and throwing

Sports: Golf, tennis, baseball, hockey

Assessment: Seated rotation, gait observation trunk counter-rotation

Exercises: Open book, lunge with rotation, cable chop/lift

Evidence: Functional rotation patterns validated kinetically

References: Myers; McGill rotational training

RECORD: Functional Lines (FL)

ID: CHN-T-005

Chain: Functional Lines (FL)

Muscles Included: Anterior: pec to opposite adductors via rectus abdominis; Posterior: lat to opposite glute via TLF

Function: Cross-body stabilization in gait and sport

Clinical Importance: Contralateral lat-glute delay in LBP and running injury

Sports: Running, throwing, cross-country skiing

Assessment: Single leg stance reach, bird dog cross-body

Exercises: Bird dog, cross-body cable, single arm row opposite leg

Evidence: Sahrmann and McGill cross-body patterns supported

References: Myers; McGill bird dog evidence

RECORD: Lateral Line (LL)

ID: CHN-T-006

Chain: Lateral Line (LL)

Muscles Included: Peroneals, ITB/TFL, glute med, QL, intercostals, SCM

Function: Lateral stability; side bending; frontal plane control

Clinical Importance: QL-thoracic lateral flexion pain; ITB-knee but thoracic contribution

Sports: Skating, basketball cutting, side plank sports

Assessment: Single leg squat frontal plane; side bend ROM

Exercises: Side plank, lateral line stretch, QL stretch side-lying

Evidence: Frontal plane training reduces injury risk meta-analyses

References: Myers; Powers frontal plane knee but lateral line trunk

9. Pathologies

RECORD: Thoracic Hyperkyphosis (Postural / Age-Related)

ID: PATH-T-001

Name: Thoracic Hyperkyphosis (Postural / Age-Related)

Region: Thoracic spine T1-T12

Epidemiology: Prevalence increases with age; adolescent postural kyphosis common desk workers

Etiology: Prolonged flexion posture, osteoporosis, Scheuermann excluded, muscle imbalance pec/upper trap dominance

Pathophysiology: Anterior soft tissue shortening; posterior extensor weakness; increased flexion moment on thoracic segments

Clinical Presentation: Rounded shoulders, forward head, stiff extension, aching mid-upper back after sitting

Red Flags: Sudden severe deformity trauma, neurological deficit, fever weight loss — not typical postural

Differential Diagnosis: Scheuermann disease, compression fracture, ankylosing spondylitis, tumor

Examination: Observe sagittal alignment; thoracic extension ROM; scapular position; breathing pattern

Special Tests: Adam's forward bend for structural vs postural; extension active/passive

Imaging: X-ray lateral if structural deformity >60 deg or progressive; DEXA if osteoporosis suspected

Conservative Management: Postural education, ergonomic setup, activity breaks, pain education

Manual Therapy: Thoracic extension mobilization PA; rib mobilization; soft tissue pec/scalene

Exercise Prescription: Extension endurance, lower trap, deep neck flexors, thoracic rotation, row patterns

Prognosis: Good response postural cases weeks-months; structural Scheuermann different course

Return to Sport: Full extension ROM sport-specific; swimming golf rotation sports need thoracic mobility

Evidence: Motor control and extension exercise Level B chronic mid-back pain

References: Katzman hyperkyphosis review; Bae thoracic posture exercise RCTs

RECORD: Scheuermann Disease (Juvenile Kyphosis)

ID: PATH-T-002

Name: Scheuermann Disease (Juvenile Kyphosis)

Region: Thoracic T7-T9 apex typical

Epidemiology: 1-8% adolescents; males > females

Etiology: Vertebral endplate irregularity; wedge vertebrae >5 deg three consecutive levels

Pathophysiology: Rigid kyphosis; anterior vertebral wedging; Schmorl nodes

Clinical Presentation: Fixed rounded back adolescent; pain activity-related; hamstring tightness common

Red Flags: Progressive deformity, neurological symptoms, severe pain night pain

Differential Diagnosis: Postural kyphosis (corrects extension), AS, infection, tumor

Examination: Rigid kyphosis on extension; positive Adam's structural curve; reduced ROM

Special Tests: Extension correction test; Adam's forward bend rib hump thoracic

Imaging: Lateral X-ray: wedge vertebrae, endplate changes, kyphosis >45 deg diagnostic

Conservative Management: PT monitoring growth; extension strengthening; bracing if progressive moderate-severe

Manual Therapy: Mobilization limited rigid segments; adjacent level mobility

Exercise Prescription: Core stability, thoracic extension isometrics, hamstring stretch, sport modification

Prognosis: Often stable after skeletal maturity; bracing if >55 deg kyphosis and growing

Return to Sport: Low-impact until pain controlled; contact sport individualized by severity

Evidence: Bracing evidence moderate for progressive curves; PT for pain function

References: Lowe Scheuermann management; Monticone exercise studies

RECORD: Thoracic Idiopathic Scoliosis

ID: PATH-T-003

Name: Thoracic Idiopathic Scoliosis

Region: Thoracic or thoracolumbar curves

Epidemiology: 2-3% adolescents screened; adults degenerative scoliosis separate entity

Etiology: Multifactorial genetic, growth-related; not postural habit alone

Pathophysiology: Vertebral rotation, lateral curvature, rib hump; asymmetric loading costovertebral joints

Clinical Presentation: Visible asymmetry shoulders/ribs; back ache; rarely cardiopulmonary if severe

Red Flags: Rapid progression prepubertal, neurological signs, bowel/bladder — rare

Differential Diagnosis: Postural scoliosis, leg length, Scheuermann, tumor syringomyelia

Examination: Adam's forward bend; scoliometer; limb length; neurological screen

Special Tests: Adam's test, trunk rotation measurement, Adams forward bend rib hump

Imaging: Full spine PA/lateral standing; Cobb angle; maturity Risser

Conservative Management: Observation mild curves; Schroth/PSSE; bracing per SRS criteria

Manual Therapy: Segmental mobility adjacent to curve; not correct structural curve manually

Exercise Prescription: Schroth 3-dimensional exercises, core, respiratory expansion concave side

Prognosis: Progression risk highest during growth spurts; adult stable curves often manageable

Return to Sport: Sport usually continued; monitor curve sport-specific asymmetry

Evidence: Bracing 45-50 deg Cobb growing skeleton evidence; PSSE growing evidence emerging

References: Negrini scoliosis exercise Cochrane; Weinstein bracing RCT

RECORD: Thoracic Compression Fracture

ID: PATH-T-004

Name: Thoracic Compression Fracture

Region: Mid-lower thoracic T6-T12 common osteoporosis

Epidemiology: Osteoporotic elderly common; trauma young athletes rare high-energy

Etiology: Axial load flexion; osteoporosis; metastatic pathologic fracture

Pathophysiology: Anterior vertebral height loss; acute or insufficiency fracture

Clinical Presentation: Localized midline pain; increased with flexion loading; height loss multiple fractures

Red Flags: Progressive neuro deficit, bowel/bladder, fever, known malignancy, high-energy incomplete assessment

Differential Diagnosis: Disc herniation thoracic rare, MSK strain, herpes zoster, aortic pathology referred

Examination: Palpation SP tenderness; flexion painful; neuro exam lower extremities reflexes

Special Tests: Percussion tenderness; single leg stance load test cautious

Imaging: X-ray AP/lateral; MRI if neuro deficit osteoporosis unclear; CT surgical planning

Conservative Management: Relative rest, analgesia medical, thoracic extension brace some protocols, early ambulation

Manual Therapy: Gentle mobilization after acute phase; avoid aggressive flexion early

Exercise Prescription: Extension progressive loading, balance, osteoporosis weight-bearing medical co-management

Prognosis: Most osteoporotic heal 8-12 weeks pain; kyphoplasty if severe pain selected cases

Return to Sport: Return after pain-free extension loading; contact sport after structural healing trauma cases

Evidence: Early mobilization superior to bed rest osteoporotic VF; EAA guidelines

References: McCarthy osteoporotic vertebral fracture management

RECORD: Thoracic Facet Joint Syndrome

ID: PATH-T-005

Name: Thoracic Facet Joint Syndrome

Region: Unilateral or bilateral facet pain T4-T10 common

Epidemiology: Middle age; degenerative; whiplash upper thoracic; rotation sports

Etiology: Facet arthropathy, synovitis, meniscoid entrapment, postural overload extension

Pathophysiology: Inflamed facet capsule; medial branch nociception

Clinical Presentation: Paraspinal pain 1-2 cm lateral to SP; extension rotation provocative; morning stiffness

Red Flags: Weight loss, night pain unrelenting, neuro deficit

Differential Diagnosis: Discogenic pain, rib dysfunction, internal organ referred, costotransverse sprain

Examination: Extension rotation reproduces; segmental hypomobility or hypermobility

Special Tests: Extension rotation test; PA palpation facet line

Imaging: MRI facet effusion limited sensitivity; diagnostic medial branch block gold standard invasive

Conservative Management: Activity modification, NSAIDs short term medical, PT mobilization

Manual Therapy: Grade III-IV PA glides, rotation mobilization, dry needling medial branch region

Exercise Prescription: Extension control, rotation mobility, scapular stability

Prognosis: Good with conservative care; RF ablation chronic cases after diagnostic block

Return to Sport: Gradual return rotation extension sports after pain-free ROM

Evidence: Thoracic RF medial branch evidence extrapolated lumbar Level B

References: Manchikanti facet joint interventions review

RECORD: Costochondritis (Tietze if swelling)

ID: PATH-T-006

Name: Costochondritis (Tietze if swelling)

Region: Ribs 2-5 costochondral/sternocostal anterior chest

Epidemiology: Common young adults; female slight predominance; no trauma required

Etiology: Overuse upper body, coughing, viral illness, repetitive strain

Pathophysiology: Inflammation costochondral junction cartilage; not infection

Clinical Presentation: Sharp anterior chest pain; reproducible palpation; worse deep breath push-up

Red Flags: Fever, diaphoresis, exertional crushing pain, tachycardia — cardiac red flags

Differential Diagnosis: Cardiac ACS, PE, pneumonia, rib fracture, Tietze with swelling

Examination: Palpation costochondral junction reproduces; normal cardiac auscultation if cleared

Special Tests: Chest wall palpation; no single special test; reproduce with push-up

Imaging: Clinical diagnosis; X-ray if trauma; exclude cardiac per protocol

Conservative Management: Reassurance, NSAIDs, activity modification, heat

Manual Therapy: Gentle rib mobilization posterior; soft tissue intercostals; avoid aggressive anterior

Exercise Prescription: Gradual push-up return, breathing exercises, pec stretch

Prognosis: Self-limiting weeks to months; recurrence possible

Return to Sport: Return when palpation pain minimal and push/pull progressions tolerated

Evidence: NSAIDs and reassurance standard; PT adjunct low evidence

References: Disla costochondritis review; Family Practice guidelines chest wall pain

RECORD: Rib Fracture (Traumatic / Stress)

ID: PATH-T-007

Name: Rib Fracture (Traumatic / Stress)

Region: Ribs 4-9 most common traumatic; 1st rib high energy

Epidemiology: Trauma MVA falls contact sports; stress rowers golfers baseball pitchers

Etiology: Direct blow; repetitive torsion; coughing severe osteoporotic

Pathophysiology: Cortical disruption; pain inspiration; protective splinting intercostals

Clinical Presentation: Point tenderness; crepi if significant; pain cough sneeze twist

Red Flags: Multiple ribs flail, pneumothorax, hemothorax, abdominal trauma association

Differential Diagnosis: MSK strain, costochondritis, referred visceral pain

Examination: Palpation along rib; inspiration pain; observe breathing shallow

Special Tests: Hooking test slipping rib different; chest expansion measurement

Imaging: X-ray often confirms; CT if complex; ultrasound point of care emerging

Conservative Management: Analgesia, pulmonary hygiene incentive spirometry, rib belt controversial

Manual Therapy: Generally avoid aggressive mobilization acute; posterior rib grade I-II after pain subsides

Exercise Prescription: Diaphragmatic breathing, gradual rotation, return sport 6-8 weeks typical uncomplicated

Prognosis: Healing 6 weeks bone; pain may persist longer intercostal neuralgia

Return to Sport: Contact sport after radiographic healing and pain-free tackle simulation

Evidence: Early mobilization and analgesia improve pulmonary outcomes trauma guidelines

References: Bemelman rib fracture management review

RECORD: Slipping Rib Syndrome (Painful Rib Hypermobility)

ID: PATH-T-008

Name: Slipping Rib Syndrome (Painful Rib Hypermobility)

Region: Ribs 8-10 false ribs anterior cartilage

Epidemiology: Underdiagnosed; athletes swimmers rowers; post-trauma

Etiology: Hypermobility costal cartilage; trauma; congenital laxity

Pathophysiology: Anterior rib tip moves excessively; intercostal nerve irritation

Clinical Presentation: Lower anterior rib pain; clicking popping; Hooking test positive

Red Flags: Abdominal pathology, cholecystitis, hepatic referral

Differential Diagnosis: Costochondritis, intercostal strain, T8-T10 radiculopathy

Examination: Hooking maneuver reproduces click pain; palpation mobile rib tip

Special Tests: Hooking test; ultrasound dynamic subluxation

Imaging: Ultrasound dynamic best; X-ray limited

Conservative Management: Rest, NSAIDs, rib support taping, activity modification

Manual Therapy: Posterior rib stabilization; core breathing; avoid aggressive anterior manipulation

Exercise Prescription: Core stability, gradual return swimming rowing

Prognosis: Conservative often successful; surgical excision refractory cases

Return to Sport: Swimming rowing after Hooking negative and core stable

Evidence: Surgical case series good outcomes refractory; conservative first line expert

References: Heiderscheit slipping rib syndrome review; McBeath diagnostic criteria

RECORD: Intercostal Muscle Strain

ID: PATH-T-009

Name: Intercostal Muscle Strain

Region: Any interspace; rotation sports common

Epidemiology: Common athletes tennis golf baseball twisting

Etiology: Sudden rotation extension; sneeze; overuse repetitive trunk rotation

Pathophysiology: Muscle fiber tear external/internal intercostal; inflammation

Clinical Presentation: Sharp lateral chest/back pain twist; palpation intercostal groove; bracing pain

Red Flags: Cardiac pulmonary if anterior crushing dyspnea

Differential Diagnosis: Rib fracture, costochondritis, thoracic radiculopathy

Examination: Palpation intercostal; reproduce rotation; resisted rotation pain

Special Tests: Rotation reproduces; intercostal palpation

Imaging: Usually none; X-ray if fracture suspected

Conservative Management: Relative rest ice early; NSAIDs short term; avoid Valsalva early

Manual Therapy: Gentle soft tissue; rib mobilization after acute; dry needling optional

Exercise Prescription: Progressive rotation, core anti-rotation, return throwing progression

Prognosis: 2-6 weeks typical grade I-II

Return to Sport: Sport-specific rotation progression when pain-free isometric rotation

Evidence: Limited RCTs; standard muscle strain protocols applied

References: Bordoni intercostal muscle anatomy and function

RECORD: Thoracic Disc Herniation

ID: PATH-T-010

Name: Thoracic Disc Herniation

Region: T8-T12 most common thoracic disc levels clinically

Epidemiology: Rare vs cervical/lumbar; ~0.25-4% disc herniations

Etiology: Degeneration; trauma; Scheuermann association

Pathophysiology: Posterolateral disc protrusion/extrusion; cord compression central paramedian

Clinical Presentation: Band-like dermatomal pain; myelopathy if central — gait disturbance, UMN signs

Red Flags: Progressive weakness, bowel bladder, saddle anesthesia — surgical emergency

Differential Diagnosis: Radiculopathy abdominal mimic, MSK strain, tumor infection

Examination: Neuro exam essential; dermatomal sensory; reflexes; Babinski if myelopathy

Special Tests: Slump test; hyperreflexia; Hoffmann upper motor neuron

Imaging: MRI gold standard; T2 disc herniation cord compression

Conservative Management: Conservative if no myelopathy; epidural selective cases; PT pain centralization approach case dependent

Manual Therapy: Gentle extension mobilization if tolerated; avoid flexion loading acute radicular

Exercise Prescription: Core stability, extension bias if centralizes, gradual loading

Prognosis: Myelopathy requires surgery; radicular may resolve conservative months

Return to Sport: No sport until neuro cleared; gradual return non-contact then contact

Evidence: Surgery for myelopathy strong; conservative radicular limited evidence case series

References: Wood thoracic disc herniation outcomes review

RECORD: Thoracic Outlet Syndrome — Neurogenic

ID: PATH-T-011

Name: Thoracic Outlet Syndrome — Neurogenic

Region: Brachial plexus compression scalene/costoclavicular/pec minor

Epidemiology: Most common TOS ~95%; women young adults; overhead athletes

Etiology: Anomaly cervical rib; posture; trauma; repetitive overhead

Pathophysiology: Plexus compression ischemic and mechanical; fibrous bands

Clinical Presentation: Ulnar paresthesia hand; neck shoulder pain; overhead intolerance

Red Flags: Progressive atrophy intrinsic hand; vascular TOS signs; Pancoast tumor

Differential Diagnosis: Cervical radiculopathy, CTS, ulnar neuropathy cubital tunnel

Examination: Roos Adson Wright battery; posture; first rib mobility; strength intrinsics

Special Tests: Roos, Adson, Wright, costoclavicular, ULTT

Imaging: X-ray cervical rib; MRI plexus; EMG/NCS chronic

Conservative Management: Posture, first rib mobilization, scalene release, strengthening lower/middle trap

Manual Therapy: Soft tissue scalenes; cervical/thoracic mobilization; nerve glides cautious

Exercise Prescription: Chin tuck, scapular retraction, diaphragmatic breathing, progressive overhead

Prognosis: 50-90% mild-moderate conservative success literature variable

Return to Sport: Overhead sport gradual throw/swim program symptom-guided

Evidence: Conservative first line Level C expert; surgery failed rehab documented vascular

References: Sanders TOS; Nord conservative outcomes

RECORD: Thoracolumbar Junction Syndrome

ID: PATH-T-012

Name: Thoracolumbar Junction Syndrome

Region: T11-L2 transition zone

Epidemiology: Common active adults; dancers gymnasts weightlifters

Etiology: Shear stress transition; facet orientation change; QL iliolumbar stress

Pathophysiology: Increased sagittal mobility lower thoracic; pars stress rare T12

Clinical Presentation: Local pain extension rotation; QL referral flank; lifting aggravates

Red Flags: Severe trauma fracture dislocation; cauda equina lower

Differential Diagnosis: Lumbar facet, disc L1-L2, renal referral

Examination: Thoracolumbar ROM; QL length strength; extension rotation testing

Special Tests: Extension rotation; one-leg standing load

Imaging: X-ray oblique if pars concern; MRI if neuro

Conservative Management: Activity modification; core lumbopelvic control; thoracic and hip mobility

Manual Therapy: T12-L1 mobilization; QL soft tissue; iliolumbar region

Exercise Prescription: Bird dog, dead bug, hip hinge, anti-rotation, gradual deadlift progression

Prognosis: Good with motor control; chronic if undiagnosed transition overload

Return to Sport: Lifting sports after pain-free hinge and loaded rotation

Evidence: Motor control training Level B chronic spinal pain

References: McGill spine stability; Sahrman movement impairment

10. Special Tests (Thoracic / Rib / Breathing)

RECORD: Thoracic Extension Rotation Test (BERT)

ID: TST-T-001

Test Name: Thoracic Extension Rotation Test (BERT)

Purpose: Provoke thoracic facet and costovertebral pain with extension rotation

Structures Stressed: Thoracic facets, costovertebral joints, paraspinal muscles

Procedure: Patient seated arms crossed; therapist extension rotation each side end-range

Positive Finding: Reproduction familiar posterior/thoracic pain; compare sides asymmetry

Sensitivity: Moderate for thoracic pain source localization

Specificity: Low specificity — multiple structures

Clinical Utility: Useful clinical reasoning tool segmental pain; not standalone diagnosis

Precautions: Avoid acute fracture, severe osteoporosis, hypermobility instability

Evidence: Clinical reasoning support; limited psychometric studies thoracic specifically

References: Manual therapy texts Cleland; Flynn clinical prediction rules thoracic

RECORD: Spring Test (Central PA Thoracic)

ID: TST-T-002

Test Name: Spring Test (Central PA Thoracic)

Purpose: Assess segmental stiffness and pain response central posterior-anterior glide

Structures Stressed: Intervertebral disc, vertebral body, ALL, segmental mobility

Procedure: Patient prone; central PA pressure each thoracic spinous process grades I-IV

Positive Finding: Pain reproduction; stiffness compared adjacent segments; muscle guarding

Sensitivity: N/A quantitative — qualitative stiffness assessment

Specificity: N/A

Clinical Utility: Fundamental manual therapy assessment; guides mobilization level

Precautions: Fracture, tumor, infection, anticoagulation — caution grade IV

Evidence: Foundation of Maitland mobilization assessment

References: Hengeveld Maitland Vertebral Manipulation

RECORD: Rib Spring Test

ID: TST-T-003

Test Name: Rib Spring Test

Purpose: Assess costotransverse/costovertebral hypomobility or pain

Structures Stressed: Costotransverse joint, costovertebral joint, intercostals

Procedure: Patient side-lying/prone; PA pressure on posterior rib angle

Positive Finding: Asymmetric spring; pain reproduction; hypomobility fixation

Sensitivity: Clinical tradition high; limited formal psychometrics

Specificity: Low — many anterior chest pain sources

Clinical Utility: Essential rib dysfunction assessment; guides rib mobilization

Precautions: Rib fracture acute, osteoporosis, tumor

Evidence: Expert consensus manual medicine

References: Greenman Principles of Manual Medicine

RECORD: First Rib Mobility Test

ID: TST-T-004

Test Name: First Rib Mobility Test

Purpose: Assess first rib elevation hypomobility TOS contribution

Structures Stressed: First rib, scalenes, costovertebral T1, brachial plexus proximity

Procedure: Supine or seated palpate first rib supraclavicular; compare side-to-side elevation spring

Positive Finding: Restricted first rib; tenderness; symptom reproduction TOS

Sensitivity: Moderate in TOS clinical series not isolated

Specificity: Low alone

Clinical Utility: Part of TOS assessment battery with Roos Adson

Precautions: Vascular TOS caution aggressive testing

Evidence: TOS literature Sanders; physiotherapy TOS protocols

References: Sanders physical diagnosis TOS

RECORD: Hooking Maneuver (Slipping Rib Test)

ID: TST-T-005

Test Name: Hooking Maneuver (Slipping Rib Test)

Purpose: Detect anterior rib hypermobility ribs 8-10

Structures Stressed: Costal cartilage false ribs, intercostal nerve

Procedure: Hook fingers under costal margin pull anterior; reproduce click pain

Positive Finding: Clicking; anterior pain reproduction; symptomatic relief after repositioning sometimes

Sensitivity: High in slipping rib confirmed cases

Specificity: Moderate — operator dependent

Clinical Utility: Key test slipping rib syndrome diagnosis

Precautions: Avoid acute fracture, post-surgical

Evidence: Heiderscheit criteria; ultrasound confirmation recommended

References: McBeath slipping rib diagnostic criteria

RECORD: Adam's Forward Bend Test (Scoliosis Screen)

ID: TST-T-006

Test Name: Adam's Forward Bend Test (Scoliosis Screen)

Purpose: Screen structural scoliosis vertebral rotation rib hump

Structures Stressed: Thoracic spine rotation, rib cage asymmetry

Procedure: Standing forward flexion; observe rib hump thoracic; scoliometer optional

Positive Finding: Rib hump rotation asymmetry; structural curve does not correct

Sensitivity: Screening test high sensitivity structural curve

Specificity: Low specificity postural

Clinical Utility: School screening standard; referral Cobb >10 deg rotation

Precautions: Non-diagnostic alone — imaging for confirmation

Evidence: SRS screening guidelines

References: Adolescent idiopathic scoliosis screening literature

RECORD: Chest Expansion Measurement

ID: TST-T-007

Test Name: Chest Expansion Measurement

Purpose: Quantify restrictive thoracic cage motion respiratory or ankylosis

Structures Stressed: Intercostals, costovertebral joints, thoracic spine

Procedure: Tape measure at xiphoid level max inspiration vs expiration difference

Positive Finding: <5 cm expansion abnormal adult; compare norms age sex

Sensitivity: Moderate restrictive disease detection

Specificity: Low musculoskeletal specificity

Clinical Utility: Ankylosing spondylitis monitoring; post-thoracic surgery

Precautions: Acute rib fracture pain limit effort

Evidence: ASAS assessment criteria includes chest expansion

References: Braun ankylosing spondylitis assessment

RECORD: Wall Angel / Flexion-Rotation Test

ID: TST-T-008

Test Name: Wall Angel / Flexion-Rotation Test

Purpose: Combined thoracic extension rotation mobility assessment

Structures Stressed: Thoracic spine, ribs, scapulothoracic

Procedure: Back to wall arms W position slide up maintaining contact lumbar/thoracic

Positive Finding: Inability maintain contact; pain; asymmetry shoulder elevation

Sensitivity: N/A — functional mobility screen

Specificity: N/A

Clinical Utility: Quick clinic screen desk workers swimmers overhead athletes

Precautions: Acute pain limit range

Evidence: Functional movement screening adaptation

References: Cook FMS influenced screens; Kibler shoulder

RECORD: Seated Rotation Active ROM Test

ID: TST-T-009

Test Name: Seated Rotation Active ROM Test

Purpose: Compare thoracic rotation bilateral sport baseline

Structures Stressed: Thoracic spine segments, rib articulations

Procedure: Seated knees fixed hands across chest rotate each way measure degrees inclinometer

Positive Finding: <30 deg or >10 deg asymmetry clinically notable many sports

Sensitivity: N/A

Specificity: N/A

Clinical Utility: Golf tennis baseball hockey screening; track rehab progress

Precautions: Acute strain perform within pain-free range

Evidence: Sport-specific normative data emerging

References: Johnson golf thoracic rotation norms

RECORD: Hi-Lo Breathing Assessment

ID: TST-T-010

Test Name: Hi-Lo Breathing Assessment

Purpose: Distinguish diaphragmatic vs apical dominant breathing

Structures Stressed: Diaphragm, scalenes, SCM, intercostals

Procedure: Supine one hand chest one abdomen; observe which rises first/more with inhale

Positive Finding: Chest rises predominantly = apical pattern; abdomen = diaphragmatic

Sensitivity: N/A qualitative

Specificity: N/A

Clinical Utility: Breathing retraining baseline; TOS neck pain cases

Precautions: None significant

Evidence: Clinical physiotherapy respiratory assessment standard

References: Chaitow breathing pattern disorders

11. Clinical Examination Template

RECORD: Thoracic Spine & Rib Cage Clinical Examination

Template Name: Thoracic Spine & Rib Cage Clinical Examination

ID: EXAM-T-001

Subjective: Onset mechanism; pain location deep vs superficial; breathing cough sneeze effect; morning stiffness; sport occupation; red flags cardiac pulmonary neuro; prior imaging surgery

Observation: Posture sagittal frontal planes; scapular position winging; rib cage symmetry; breathing pattern apical vs diaphragmatic; skin marks

Palpation: Spinous processes TP paraspinals; costotransverse angles; costochondral junctions if anterior; facet line; muscle tone scalenes pec

Active ROM: Thoracic flexion extension rotation lateral flexion seated; combined patterns sport-specific

Passive ROM: Segmental PA glides; rib spring; first rib mobility

Special Tests: As indicated: Adam's test, Roos TOS, Hooking rib, extension rotation, slump if radicular

Neurological: Dermatomes T1-T12 key points; myotomes; reflexes if indicated; upper motor neuron screen myelopathy

Functional: Push-up plus; wall angel; overhead reach; lift simulation

Outcome Measures: Numeric pain rating; Oswestry if thoracolumbar overlap; SRS-22 scoliosis; chest expansion

Clinical Impression: Synthesize mobility vs stability; regional interdependence cervical lumbar shoulder; breathing contribution

Plan: Education; manual therapy grade; exercise prescription phase; referral imaging medical if red flags

References: ISIS guidelines; clinical reasoning textbooks

RECORD: Thoracic Red Flags & Medical Referral Screen

Template Name: Thoracic Red Flags & Medical Referral Screen

ID: EXAM-T-002

Cardiac: Exertional chest pain; diaphoresis; radiation jaw/arm; dyspnea disproportionate; syncope — emergency cardiac workup before MSK treatment

Pulmonary: Fever cough hemoptysis; unexplained weight loss; pleuritic pain tachypnea — pulmonary infection PE tumor workup

Neurological: Progressive bilateral leg weakness; gait disturbance; bowel bladder dysfunction; saddle anesthesia — cord compression emergency MRI

Oncologic: Age >50 new unremitting pain; night pain unrelieved rest; history cancer; unexplained weight loss — imaging and medical referral

Infection: Fever IV drug use immunocompromise recent procedure — MRI contrast discitis epidural abscess

Trauma: High-energy mechanism; suspected fracture dislocation; neuro deficit post-trauma — imaging trauma protocol

Vascular TOS: Arm swelling cyanosis acute effort thrombosis — emergency vascular workup

Fracture Risk: Known osteoporosis acute flexion load severe pain — imaging before aggressive manual therapy

References: NICE red flags; clinical reasoning textbooks; emergency medicine guidelines

12. Rehabilitation Phases

RECORD: Rehab Phase 1 — Acute Pain Control (Thoracic)

ID: REH-T-001

Phase: Acute (0-2 weeks typical MSK)

Goals: Reduce pain/inflammation; protect tissue; maintain safe ROM; patient education red flags

Pain Management: Relative rest from aggravators; heat/ice preference; medical analgesia coordinated

Mobility: Pain-free gentle rotation flexion extension within tolerance; diaphragmatic breathing

Strengthening: Isometric scapular setting; gentle core activation dead bug level 1

Manual Therapy: Grade I-II mobilization; soft tissue gentle; rib grade I if rib involvement

Modalities: TENS optional; not primary; avoid passive-only approach

Patient Education: Prognosis reassurance MSK; ergonomics; avoid prolonged flexion sitting

Progression Criteria: Pain trending down; sleep improved; basic ADL without sharp pain

References: NICE back pain principles applied thoracic MSK

RECORD: Rehab Phase 2 — Mobility Restoration

ID: REH-T-002

Phase: Subacute (2-6 weeks)

Goals: Restore thoracic rotation extension; normalize breathing; segmental mobility

Pain Management: Gradual exposure to previously painful movements

Mobility: Open book, thread needle, foam roller extension, cat-camel, rib expansion

Strengthening: Band rows, scapular retraction endurance, prone Y/T low load

Manual Therapy: Grade III-IV PA glides hypomobile segments; costovertebral mobilization

Modalities: Reduce emphasis; active care primary

Patient Education: Importance movement; desk breaks; sleeping support

Progression Criteria: ROM >75% baseline; pain <3/10 daily activities

References: Thoracic manipulation chronic neck pain indirect evidence

RECORD: Rehab Phase 3 — Strengthening & Motor Control

ID: REH-T-003

Phase: Strengthening (4-10 weeks)

Goals: Paraspinal endurance; scapulothoracic control; anti-rotation stability

Pain Management: Manage post-exercise soreness vs pain flare distinction

Mobility: Maintain rotation extension; sport-specific patterns

Strengthening: Rows, face pulls, lat pull, bird dog, pallof press, dead bug progressions

Manual Therapy: Maintenance hypomobile segments; soft tissue as needed

Modalities: Minimal

Patient Education: Load management; technique lifting sport

Progression Criteria: Strength endurance targets met; scapular dyskinesis improved

References: McGill big three; Kibler scapular protocols

RECORD: Rehab Phase 4 — Return to Sport / Function

ID: REH-T-004

Phase: Return to Sport (variable 8-16 weeks)

Goals: Sport-specific loads; rotation power; contact tolerance if applicable

Pain Management: Pain-free sport progressions; flare management plan

Mobility: Full sport-required ROM symmetric

Strengthening: Plyometric rotation; overhead progressions; weighted carries

Manual Therapy: Maintenance only

Modalities: None routine

Patient Education: Warm-up routine; maintenance exercises; recognize early warning signs

Progression Criteria: Sport test battery passed; confidence; coach clearance

References: Sport-specific RTS criteria individualized

RECORD: Rehab Phase — Post-Thoracic Surgery (General)

ID: REH-T-005

Phase: Post-surgical (protocol surgeon-specific)

Goals: Wound healing; pulmonary hygiene; gradual mobility per surgical restrictions

Pain Management: Surgeon analgesia protocol; splinting incision coughing technique

Mobility: Only within surgeon restrictions — often limited rotation/extension early

Strengthening: Delayed until clearance — typically isometric then progressive

Manual Therapy: Not over incision site early; rib/thoracic distant segments gentle later

Modalities: Per hospital protocol

Patient Education: Strict adherence restrictions; incentive spirometry; signs infection

Progression Criteria: Surgeon follow-up clearance each phase

References: Institutional post-thoracotomy/ spinal surgery protocols

13. Exercise Library

RECORD: Open Book (Thoracic Rotation)

ID: EX-T-001

Name: Open Book

Region: Thoracic rotation

Purpose: Restore thoracic rotation; reduce reliance on lumbar/cervical compensation

Starting Position: Side-lying hips/knees flexed 90 deg arms extended together

Execution: Rotate top arm and trunk open like book following hand with eyes; exhale end-range

Dosage: 2-3 sets 10-15 reps each side; hold 2-3 sec end-range

Progression: Add foam roller under top knee; increase range; add light dumbbell hand

Regression: Smaller range; pillow between knees

Contraindications: Acute rib fracture; severe osteoporosis HVLA adjacent

Cues: Keep knees stacked; rotate from mid-back not just shoulder

Evidence: Commonly prescribed; indirect evidence thoracic rotation benefits throwing

References: TPI golf mobility; physiotherapy clinical practice

RECORD: Thread the Needle

ID: EX-T-002

Name: Thread the Needle

Region: Thoracic rotation

Purpose: Combined rotation extension mobility quadruped

Starting Position: Quadruped neutral spine

Execution: Slide one arm under opposite arm rotating thoracic; hip stays level

Dosage: 2-3 x 8-12 each side

Progression: Hold end-range 5 sec; band resistance

Regression: Hand on chair seated variant

Contraindications: Wrist pain quadruped; acute disc radicular flexion intolerance

Cues: Rotate through mid-back; breathe into upper back

Evidence: Clinical standard mobility exercise

References: Movement system impairment approach

RECORD: Foam Roller Thoracic Extension

ID: EX-T-003

Name: Foam Roller Extension

Region: Mid-thoracic extension

Purpose: Mobilize extension mid-thoracic counter kyphotic posture

Starting Position: Supine foam roller perpendicular mid-thoracic; hands support head

Execution: Gentle extension over roller segment by segment; avoid lumbar

Dosage: 1-2 min total; 5-10 slow extensions

Progression: Arms overhead; narrow base

Regression: Towel roll smaller diameter

Contraindications: Osteoporosis severe; acute compression fracture; dizziness

Cues: Small segment focus; exhale into extension

Evidence: Improves extension ROM short-term studies

References: Mid-back pain extension mobilization literature

RECORD: Cat-Camel (Segmental)

ID: EX-T-004

Name: Cat-Camel

Region: Full spine thoracic emphasis

Purpose: Segmental flexion-extension awareness; spinal warm-up

Starting Position: Quadruped

Execution: Alternate flexion rounding and extension sagging; segmental initiation cue

Dosage: 2 x 10-15 slow repetitions

Progression: Pause end ranges; eyes closed proprioception

Regression: Seated cat-camel

Contraindications: Acute radicular flexion/extension intolerance

Cues: Initiate from mid-thoracic not lumbar only

Evidence: McGill spine warm-up standard

References: McGill Ultimate Back Fitness

RECORD: Prone Y Raise (Lower Trap)

ID: EX-T-005

Name: Prone Y

Region: Scapulothoracic / lower trapezius

Purpose: Lower trap activation; upward rotation posterior tilt

Starting Position: Prone arms Y ~130 deg abduction thumbs up

Execution: Lift arms squeezing lower trap; minimal upper trap

Dosage: 3 x 10-15 holds 3 sec; light weight 0.5-2 kg

Progression: Incline bench; add weight

Regression: Smaller angle; no weight

Contraindications: Acute shoulder impingement painful

Cues: Slide scapula into pocket; long neck

Evidence: High lower trap EMG prone Y

References: Reinold scapular EMG exercises

RECORD: Wall Slide / Wall Angel

ID: EX-T-006

Name: Wall Angel

Region: Thoracic extension scapular

Purpose: Combined extension scapular upward rotation mobility

Starting Position: Back against wall feet slightly forward arms W

Execution: Slide arms up maintaining wrist/elbow/back contact if possible

Dosage: 2-3 x 8-10 slow

Progression: Add mini band; single arm

Regression: Partial range acceptable

Contraindications: Shoulder pain end-range

Cues: Ribs down; breathe lateral expansion

Evidence: Functional screen and rehab common

References: Kibler shoulder rehabilitation

RECORD: Bird Dog

ID: EX-T-007

Name: Bird Dog

Region: Thoracolumbar stability

Purpose: Anti-extension anti-rotation core; multifidus erector coactivation

Starting Position: Quadruped neutral

Execution: Extend opposite arm leg maintaining trunk level hold

Dosage: 3 x 10 each side 5-10 sec hold

Progression: Band resistance; unstable surface

Regression: Arm only or leg only

Contraindications: Wrist pain; acute extension pain

Cues: Level hips; long spine

Evidence: McGill endurance protocol evidence

References: McGill bird dog studies

RECORD: Pallof Press Anti-Rotation

ID: EX-T-008

Name: Pallof Press

Region: Thoracic anti-rotation

Purpose: Rotational stability; oblique activation; sport transfer

Starting Position: Standing band/cable at chest height

Execution: Press band forward resisting rotation; hold extended

Dosage: 3 x 10-15 each side

Progression: Kneeling half-kneeling split stance sport stance

Regression: Lighter band shorter hold

Contraindications: Acute intercostal strain painful rotation

Cues: Ribs stacked over pelvis; exhale on press

Evidence: High anti-rotation demand EMG obliques

References: McGill anti-rotation training

RECORD: Diaphragmatic Breathing 90-90

ID: EX-T-009

Name: 90-90 Diaphragmatic Breathing

Region: Respiratory / core

Purpose: Restore diaphragm dominance; reduce apical pattern

Starting Position: Supine hips/knees 90 deg on chair/wall

Execution: Inhale expand lower ribs/back; exhale ribs down without chest lift

Dosage: 5-10 min daily; 5-8 breaths/min

Progression: Add arm movement; standing

Regression: Crocodile breathing prone

Contraindications: None typical

Cues: 360 expansion; slow nasal inhale

Evidence: Breathing retraining chronic pain Level B

References: Chaitow; Kolar DNS breathing

RECORD: Side-Lying Rib Expansion

ID: EX-T-010

Name: Rib Expansion Side-Lying

Region: Intercostals costovertebral

Purpose: Mobilize restricted hemithorax; post-rib injury; scoliosis convex side

Starting Position: Side-lying restricted side up or down per protocol

Execution: Inhale into top/back ribs; manual facilitation optional

Dosage: 2-3 min each side

Progression: Add arm overhead

Regression: Seated

Contraindications: Acute rib fracture painful side

Cues: Breathe into hand placement lateral ribs

Evidence: Schroth scoliosis breathing principles

References: Schroth PSSE methods

RECORD: Serratus Wall Slide

ID: EX-T-011

Name: Serratus Wall Slide

Region: Scapulothoracic protraction upward rotation

Purpose: Serratus activation; scapular protraction control

Starting Position: Forearms on wall slide position

Execution: Slide arms up maintaining protraction plus at top

Dosage: 3 x 10-12

Progression: Foam roller wall; single arm

Regression: Partial range

Contraindications: Long thoracic nerve acute winging painful

Cues: Push wall away; plus at top

Evidence: High serratus EMG wall slide variations

References: Decker serratus EMG

RECORD: Quadruped Serratus Push (Push-Up Plus)

ID: EX-T-012

Name: Push-Up Plus

Region: Serratus scapular

Purpose: End-range protraction serratus strength

Starting Position: Push-up position or wall incline

Execution: At top protract scapula further rounding upper back slightly

Dosage: 3 x 12-20

Progression: Full push-up; band around back

Regression: Wall incline high angle

Contraindications: Wrist pain

Cues: Separate scapula at top; do not bend elbows for plus

Evidence: Standard serratus strengthening exercise

References: Reinold push-up plus EMG

RECORD: Seated Row / Band Row

ID: EX-T-013

Name: Seated Row

Region: Scapular retraction thoracic posture

Purpose: Middle trap rhomboid endurance; posterior chain

Starting Position: Seated band or cable chest height

Execution: Pull elbows back squeeze scapulae; control return

Dosage: 3 x 12-15 moderate load

Progression: Single arm; unstable surface

Regression: Light band high reps

Contraindications: Acute shoulder impingement

Cues: Chest tall; ribs down

Evidence: High middle trap rhomboid EMG

References: Cools scapular muscle EMG

RECORD: Cable Chop / Lift (Spiral Line)

ID: EX-T-014

Name: Cable Chop/Lift

Region: Thoracic rotation obliques

Purpose: Rotational power stability; spiral line integration

Starting Position: Standing cable high-to-low chop or low-to-high lift

Execution: Rotate through thoracic hips; control eccentric

Dosage: 3 x 8-12 each direction

Progression: Heavier load; sport stance

Regression: Light band

Contraindications: Acute disc radicular rotation

Cues: Power from hips trunk follows

Evidence: Functional training rotational athletes

References: McGill rotational training guidelines

RECORD: Dead Bug

ID: EX-T-015

Name: Dead Bug

Region: Core respiratory coordination

Purpose: IAP maintenance limb loading; diaphragm-pelvic floor coordination

Starting Position: Supine 90-90 arms up

Execution: Lower opposite arm/leg maintaining neutral spine and breathing

Dosage: 3 x 8-10 each side slow

Progression: Band resistance; lower starting position

Regression: Heel taps only

Contraindications: Acute flexion intolerance

Cues: Ribs down; exhale on extension

Evidence: Core stability literature support

References: Kolar DNS; McGill modified curl-up progressions

14. Return to Sport

RECORD: RTS — General Thoracic MSK

ID: RTS-T-001

Sport/Context: General athletic population

Criteria: Pain-free ADL; symmetric rotation >45 deg; extension without pain; no night pain

Strength: Row and push patterns symmetric 90% limb symmetry optional

Functional Tests: Sport-specific movements pain-free; push-up plus; overhead reach

Progression: Linear load increase 10-15% weekly if symptom-free

Timeline: 2-8 weeks mild strain; 8-12 weeks fracture; variable surgery

Red Flags Stop: Radiating pain; numbness; chest pain cardiac; fever

References: Clinical consensus RTS spinal MSK

RECORD: RTS — Throwing / Overhead Athlete

ID: RTS-T-002

Sport/Context: Baseball tennis volleyball swimming

Criteria: Full painless overhead ROM; scapular dyskinesis resolved; TOS tests negative

Strength: Serratus lower trap endurance; rotator cuff balance; Kibler scapular tests

Functional Tests: Throwing interval program; serve progression; swim volume build

Progression: Structured interval throwing ASMI model; 10% volume rule

Timeline: 4-12 weeks post thoracic strain; longer post rib fracture

Red Flags Stop: Vascular symptoms; winging progression; ulnar paresthesia

References: ASMI interval throwing; Reinold overhead athlete RTS

RECORD: RTS — Contact / Collision Sport

ID: RTS-T-003

Sport/Context: Rugby football hockey martial arts

Criteria: Radiographic healing if fracture; pain-free contact simulation; rib cage expansion normal

Strength: Trunk rotation power; tackle bag tolerance

Functional Tests: Progressive contact drills non-contact → limited → full

Progression: Medical clearance; 6-12 weeks post rib fracture typical

Timeline: Minimum 6 weeks uncomplicated rib fracture often cited

Red Flags Stop: Flail chest history; pneumothorax; unhealed fracture pain

References: Bemelman rib fracture return contact

RECORD: RTS — Rotational Sport (Golf/Tennis/Hockey)

ID: RTS-T-004

Sport/Context: High thoracic rotation demand

Criteria: Rotation ROM symmetric within 10 deg sport norm; no pain ball contact

Strength: Rotational power chop/lift; hip-thoracic dissociation

Functional Tests: Gradual swing progression tee → half → full

Progression: Volume before intensity; monitor next-day stiffness

Timeline: 2-6 weeks intercostal; 8+ weeks significant rib injury

Red Flags Stop: Catch pain mid-swing; intercostal sharp pain

References: TPI golf return; tennis medicine protocols

RECORD: RTS — Respiratory / Endurance Sport

ID: RTS-T-005

Sport/Context: Running cycling rowing swimming distance

Criteria: Normal breathing pattern; chest expansion; no pain deep inspiration sustained

Strength: Intercostal endurance; diaphragmatic efficiency

Functional Tests: Gradual volume build 10% rule; talk test

Progression: Return to volume before intensity intervals

Timeline: Post costochondritis 2-4 weeks; post rib fracture longer

Red Flags Stop: Dyspnea disproportionate; chest pain cardiac pattern

References: Respiratory rehab post COVID applicable principles

RECORD: RTS — Weightlifting / Powerlifting

ID: RTS-T-006

Sport/Context: Squat deadlift bench overhead press

Criteria: Pain-free hip hinge; thoracic extension under load; Valsalva tolerance if used

Strength: Progressive loading 50-60-70-80-90% 1RM protocol

Functional Tests: Technique video; belt optional; no thoracic rounding pain

Progression: Return deadlift before heavy squat if extension sensitive

Timeline: Thoracolumbar strain 2-6 weeks; fracture 12+ weeks

Red Flags Stop: Radicular symptoms under load; thoracic sharp pain flexion

References: McGill spine loading guidelines lifters

15. Imaging

RECORD: Thoracic X-Ray AP/Lateral

ID: IMG-T-001

Modality: Radiography

Indications: Trauma, scoliosis screen, kyphosis assessment, suspected fracture, pre-operative

Views: PA/AP standing; lateral; optional oblique pars if indicated

Findings: Alignment Cobb kyphosis; vertebral height; rib fractures; pedicle integrity

Limitations: Low soft tissue contrast; early metastases missed; disc herniation not seen

Radiation: Moderate; ALARA principle; pregnancy shielding

Clinical Action: Correlate exam; MRI if neuro deficit normal X-ray

References: ACR appropriateness criteria thoracic spine

RECORD: Thoracic MRI

ID: IMG-T-002

Modality: MRI

Indications: Cord compression, disc herniation, infection tumor, persistent radicular/myelopathic

Views: Sagittal T1 T2 STIR; axial affected levels; contrast if infection tumor

Findings: Disc herniation; cord signal; paraspinal abscess; Modic endplate changes

Limitations: Cost; claustrophobia; implants; incidental degenerative changes

Radiation: None ionizing; gadolinium caution renal

Clinical Action: Surgical consult myelopathy; PT radicular without progressive deficit often

References: ACR MRI spine appropriateness

RECORD: CT Thoracic Spine

ID: IMG-T-003

Modality: CT

Indications: Complex fracture surgical planning; osseous detail; trauma FAST complement

Views: Axial bone window; sagittal/coronal reconstructions

Findings: Fracture comminution; canal compromise osseous; facet fracture

Limitations: Radiation higher; soft tissue inferior MRI

Radiation: Higher than X-ray; minimize pediatric pregnancy

Clinical Action: Trauma ortho/spine consult unstable patterns

References: Trauma imaging guidelines

RECORD: Bone Scan / PET

ID: IMG-T-004

Modality: Nuclear medicine

Indications: Metastatic workup; occult fracture; infection inflammatory

Views: Whole body or limited thoracic

Findings: Increased uptake fracture metastasis infection

Limitations: Non-specific; anatomical detail poor without SPECT/CT fusion

Radiation: Radiotracer exposure

Clinical Action: Oncology referral if metastases; correlate MRI

References: Oncologic imaging guidelines

RECORD: Ultrasound — Rib/Intercostal/Slipping Rib

ID: IMG-T-005

Modality: Ultrasound

Indications: Slipping rib dynamic; intercostal tear; point of care rib fracture

Views: Dynamic anterior rib margin; intercostal muscle

Findings: Hypermobility slipping rib; hematoma strain; cortical step fracture

Limitations: Operator dependent; obesity limit; posterior ribs difficult

Radiation: None

Clinical Action: Confirms slipping rib pre-surgical consult

References: Heiderscheit ultrasound slipping rib

RECORD: DEXA Scan

ID: IMG-T-006

Modality: Dual-energy X-ray absorptiometry

Indications: Osteoporosis screen compression fracture age >65 women >70 men risk factors

Views: Lumbar hip; sometimes thoracic lateral vertebral assessment FRAX

Findings: T-score osteoporosis osteopenia

Limitations: Does not predict all fractures; artifact post-surgery

Radiation: Very low

Clinical Action: Medical management bisphosphonate etc per guidelines; PT weight-bearing

References: NOF osteoporosis guidelines

RECORD: Thoracic CT Angiography / MR Angiography

ID: IMG-T-007

Modality: Angiography CT/MR

Indications: Suspected arterial/venous TOS; aortic pathology exclusion referred pain

Views: Subclavian artery vein course; cervical rib

Findings: Stenosis occlusion aneurysm cervical rib

Limitations: Contrast allergy renal; not first line MSK chest wall pain

Radiation: CTA radiation

Clinical Action: Vascular surgery TOS; emergency if acute arterial

References: Illig TOS imaging

RECORD: Scoliosis Full Spine EOS/Long Cassette

ID: IMG-T-008

Modality: Long cassette/EOS low-dose

Indications: Scoliosis monitoring Cobb; leg length; sagittal balance

Views: PA standing full spine lateral

Findings: Cobb angle; Risser; kyphosis lordosis balance

Limitations: Radiation cumulative serial imaging; EOS reduces dose

Radiation: Manage serial exposure adolescents

Clinical Action: SRS bracing surgery thresholds Cobb progression

References: SRS scoliosis imaging standards

16. Evidence & Guidelines

RECORD: Evidence — Thoracic Manipulation Neck Pain

ID: EVD-T-001

Topic: Cervicothoracic manipulation for neck pain

Evidence Level: Level B

Summary: Thoracic spine thrust manipulation may reduce neck pain and improve ROM indirectly

Clinical Application: Consider for neck pain patients with thoracic hypomobility after assessment

Reference Note: Cross et al thoracic manipulation neck pain systematic review

RECORD: Evidence — Scapular Exercise Impingement

ID: EVD-T-002

Topic: Scapular-focused exercise shoulder impingement

Evidence Level: Level B

Summary: Scapular stabilization exercises reduce impingement symptoms medium term

Clinical Application: Lower trap serratus progressions for overhead athletes

Reference Note: Struyf scapular exercise review

RECORD: Evidence — Breathing Retraining Chronic Pain

ID: EVD-T-003

Topic: Diaphragmatic breathing chronic pain

Evidence Level: Level B

Summary: Breathing retraining reduces pain anxiety in chronic pain populations

Clinical Application: Apical pattern correction TOS neck thoracic pain

Reference Note: Hopper breath therapy chronic pain

RECORD: Evidence — Schroth Scoliosis Exercise

ID: EVD-T-004

Topic: PSSE/Schroth adolescent scoliosis

Evidence Level: Level C emerging B

Summary: Physiotherapeutic scoliosis-specific exercises may reduce progression when combined bracing

Clinical Application: Refer certified Schroth therapist growing adolescents

Reference Note: Negrini Cochrane scoliosis exercise

RECORD: Evidence — Osteoporotic Vertebral Fracture Mobilization

ID: EVD-T-005

Topic: Early mobilization osteoporotic VF

Evidence Level: Level B

Summary: Early activity superior bed rest; targeted exercise reduces future fracture risk with medical therapy

Clinical Application: Gradual extension loading post VF medical clearance

Reference Note: Giangregorio exercise osteoporosis guidelines

RECORD: Evidence — TOS Conservative Management

ID: EVD-T-006

Topic: Conservative nTOS

Evidence Level: Level C

Summary: Posture exercise scalene management successful many mild-moderate cases before surgery

Clinical Application: First rib mobilization scapular strengthening program 3-6 months trial

Reference Note: Nord conservative TOS outcomes

RECORD: Evidence — Myofascial Release Thoracic

ID: EVD-T-007

Topic: Myofascial release thoracic pain

Evidence Level: Level C

Summary: Short-term ROM pain benefit; long-term superiority not established vs exercise

Clinical Application: Adjunct not replacement active exercise

Reference Note: Ajimsha myofascial release systematic review

RECORD: Evidence — Motor Control Chronic Spinal Pain

ID: EVD-T-008

Topic: Motor control exercise thoracic/lumbar

Evidence Level: Level B

Summary: Motor control training reduces chronic spinal pain vs minimal intervention

Clinical Application: Bird dog dead bug progressive lumbopelvic control

Reference Note: Saragiotto motor control Cochrane

RECORD: Evidence — Return to Sport Rib Fracture

ID: EVD-T-009

Topic: Rib fracture RTS timelines

Evidence Level: Level C expert

Summary: 6 weeks minimum uncomplicated; contact later; pulmonary complications reduced early mobilization

Clinical Application: Individualize by fracture number pain sport

Reference Note: Bemelman rib fracture management

RECORD: Evidence — Thoracic Disc Herniation Surgery

ID: EVD-T-010

Topic: Thoracic disc surgery indications

Evidence Level: Level B surgical

Summary: Myelopathy surgical decompression; radicular conservative trial reasonable

Clinical Application: Urgent MRI progressive neuro deficit

Reference Note: Wood thoracic disc review

RECORD: Evidence — Costochondritis Management

ID: EVD-T-011

Topic: Costochondritis treatment

Evidence Level: Level C

Summary: NSAIDs reassurance activity modification mainstay; PT adjunct

Clinical Application: Rule out cardiac before MSK diagnosis

Reference Note: Disla costochondritis review

RECORD: Evidence — Thoracic Kyphosis Exercise

ID: EVD-T-012

Topic: Exercise for thoracic hyperkyphosis elderly

Evidence Level: Level B

Summary: Progressive extension strengthening improves kyphosis angle and function meta-analyses

Clinical Application: Extension endurance prone rows face pulls elderly fall prevention overlap

Reference Note: Katzman hyperkyphosis exercise meta-analysis